

Self-Other Generalisation Shapes Social Interaction and Is Disrupted in Borderline Personality Disorder

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Joseph M Barnby , Jen Nguyen, Julia Griem, Magdalena Wloszek, Henry Burgess, Linda Richards, Jessica Kingston, Gavin Cooper, London Personality and Mood Disorders Consortium, P Read Montague, Peter Dayan, Tobias Nolte, Peter Fonagy

Department of Psychology, Royal Holloway, University of London, London, United Kingdom • School of Psychiatry and Clinical Neuroscience, University of Western Australia, Perth, Australia • Department for Clinical, Educational, and Health Psychology, Division of Psychology and Language Sciences, University College London, London, United Kingdom • Anna Freud, London, United Kingdom • Department of Neuroscience, Washington University in St. Louis, St. Louis, United States • Centre for Human Neuroscience Research, Virginia Tech, Roanoke, United States • Max Planck Institute of Biological Cybernetics, Tübingen, Germany • University of Tübingen, Tübingen, Germany

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The findings are **important** and intriguing, with theoretical or practical implications beyond a single subfield. The computational methods employed are clever and sophisticated and the strength of evidence is **convincing**. Many of the methodological concerns raised after the first round of review were addressed in the revised version, although all three reviewers also highlighted that the exploratory nature of the paper and the lack of clarity regarding the hypotheses make it hard to assess the impact of the results on existing theories.

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Abstract

Generalising information from ourselves to others, and others to ourselves allows for both a dependable source of navigation and adaptability in interpersonal exchange. Disturbances to social development in sensitive periods can cause enduring and distressing damage to lasting healthy relationships. However, identifying the mechanisms of healthy exchange has been difficult. We introduce a theory of self-other generalisation tested with data from a three-phase social value orientation task – the Intentions Game. We involved individuals with (n=50) and without (n=53) a diagnosis of borderline personality disorder and assessed whether infractions to self-other generalisation may explain interpersonal (in)stability. Healthy controls initially used their preferences to predict others and were influenced by their partners, leading to self-other convergence. In contrast, individuals with borderline personality disorder maintained distinct self-other representations, generating a new neutral prior when learning about others. This allowed for equal predictive performance compared to controls despite reduced updating sensitivity. Furthermore, we explored theory-driven

individual differences underpinning contagion. Overall, the findings provide a clear explanation of how self-other generalisation constrains and assists learning and how childhood adversity is associated with separation of internalised beliefs. Our model makes clear causal predictions about the mechanisms of social information generalisation concerning both joint and individual reward.

Highlights

- Humans use self-to-other transfer to constrain initial predictions about the social behaviour of others
- Information is transferred from other-to-self following observation, calibrated by the precision of beliefs
- Joint vs individualistic reward is prioritised when learning about others.
- Those diagnosed with BPD do not engage in self-other information transfer, instead keeping self and other representationally distinct.
- Higher reported childhood trauma, paranoia, and poorer trait mentalizing are associated with diminished other-to-self information transfer.

Introduction

Social animals have evolved sophisticated mechanisms for cooperation. To coordinate, individuals exchange information, enabling independent and group regulation (Emerson, 1956 [↗](#); Wheeler, 1911 [↗](#)). Interaction provides the self with insights about others and, simultaneously, about its own state within the environment. In humans, this psychological differentiation begins in utero (Ciaunica et al., 2021 [↗](#)) and continues into adulthood. This prolonged differentiation facilitates the distinction between self and others, allowing grounded social orientation and flexible adaptation. Early psychological theorists highlighted the importance of healthy social exchange in humans for integrating constructive relational blueprints, noting that disruptions during sensitive periods can impair the formation of safe social bonds (e.g., Fairbairn, 1952; 1994 [↗](#)) and foster less adaptable interpersonal beliefs (Young et al., 2006 [↗](#)).

At the individual level, humans must manage information exchange to navigate and reduce relational uncertainty. When faced with external uncertainty about others' characteristics, prior knowledge can swiftly and effectively guide predictions. Indeed, to reduce uncertainty about others, theories of the relational self (Anderson & Chen, 2002 [↗](#)) suggest that the self is the most extensive and well-grounded representation available, leading to a readily accessible initial belief that can be projected or integrated when learning about others (self-insertion; Allport, 1924 [↗](#); Kreuger & Clement, 1994 [↗](#)). When uncertainty arises regarding one's own internal state, external cues can provide anchoring information to support self-through-other calibration. Therefore, individuals can generalize information from others to themselves. contagion has been observed across economic (Suzuki et al., 2016 [↗](#)), intertemporal (Garvert et al., 2015 [↗](#)), effort discounting (Devaine & Daunzieu, 2017), and moral choice behaviour (Yu et al., 2021 [↗](#)). Social contagion is hypothesised to facilitate adaptation to social groups (Deutsch & Gerard, 1955 [↗](#); Toelch & Dolan, 2015 [↗](#)), and as a crucial component of interpersonal cohesion that relies on trust (Frith & Frith, 2012 [↗](#)).

Insights into healthy self-other relations can be gained from their frequent absence, a common and distressing reality for those with a diagnosis of Borderline Personality Disorder (BPD). BPD is characterized by interpersonal sensitivity and instability, emotional dysregulation, impulsivity, paranoia, and severe fear of abandonment (e.g., Gunderson et al., 2018; Euler et al., 2021). It is strongly associated with early childhood adversity, such as psychological, physical, and sexual abuse or neglect (Afifi et al., 2011; Bateman et al., 2023), and inconsistent parenting experiences (Crawford et al., 2009), particularly during sensitive developmental periods (Fonagy & Bateman, 2008). Early adversity interacts with pre-existing variations in stress-homeostasis mechanisms, exacerbating interpersonal disruptions (Pratt et al., 2017), underpinned by difficulties in mentalizing, leading to disruptions in social learning (e.g., Fonagy & Luyten, 2009; Nolte et al., 2023) and representations of self and other (Hanegraaf et al., 2021). The mechanisms connecting early life adversity and relational instability are elusive, making it harder to find a path to rehabilitation.

Critical questions persist concerning both typical and disrupted interaction: What computational mechanisms support efficient generalisation from self-to-other and other-to-self? Do processes operate globally or locally based on outcome (e.g. individual vs. joint reward)? Can such formal models explain observations of social learning changes in diagnoses like BPD? Understanding the processes and representations humans use in social exchange has broad implications for theories of development, group cohesion, reputation management, interpersonal synchrony, and the breakdown of social bonds.

Formal models of healthy (non-clinical) exchange posit forms of interaction between self and other representations which can reduce uncertainty during learning (whilst potentially introducing biases). As such, self-insertion accounts demonstrate that integrating self-preferences into initial predictions about others causally increases reaction times and can increase predictive accuracy when predictions are correct (Tarantola et al., 2017; Barnby et al., 2022). Social contagion has been framed as convergence to a common latent state shared by oneself and another, calibrated by self and other uncertainty (Moutoussis et al., 2016; Thomas et al., 2022). By comparison, in mental ill-health, computational models have almost exclusively probed social processes during observational learning without fully integrating self and other into future beliefs. The BPD phenotype has been associated with a potential over-reliance on social versus internal cues (Henco et al., 2020), ‘splitting’ of social latent states that encode beliefs about others (Story et al., 2023), negative appraisal of interpersonal experiences with heightened self-blame (Mancinelli et al., 2024), inaccurate inferences about others’ irritability (Hula et al., 2018), and reduced belief adaptation in social learning contexts (Siegel et al., 2020). Associative models have also been adapted to characterize ‘leaky’ self-other reinforcement learning (Ereira et al., 2018), finding that those with BPD *overgeneralize* (leak updates) about themselves to others (Story et al., 2024). Altogether, there is currently a gap in the direct causal link between insertion, contagion, and learning (in)stability.

Our present work sought to achieve two primary goals. 1. Extend prior computational theories to formalise and test the interrelation between self-insertion and social contagion in learning and choice to probe self-other generalisation, and 2. Test whether previous computational findings of differences in social learning in people with BPD can be explained by disruptions to self-other generalisation. Here, our formal hypotheses concerning self-other generalisation (Barnby et al., 2023) make clear *a priori* predictions about its relationship with observational learning that can be empirically interrogated. Exploring these factors within models can uncover algorithmic biases that healthy volunteers and people with BPD either do or do not share. We accomplish these goals using a dynamic, sequential economic paradigm called the Intentions Game. The Intentions Game builds upon the well-known Social Value Orientation framework (Murphy & Ackerman, 2011), which captures innate motivational variation in reward allocation to self and other.

We discover that healthy participants employ a mixed process of self-insertion and contagion to predict and align with the beliefs of their partners. In contrast, age, sex, and education, and social deprivation matched individuals with BPD exhibit distinct, disintegrated representations of self and other, despite showing similar average accuracy in their predictions about partners. Our model and data suggest that the previously observed computational characteristics in BPD arise from a failure of information transfer, first from self to other, and then from other to self, an *under* rather than *over* generalisation. By integrating separate computational findings, we provide a foundational model and a concise, dynamic paradigm to investigate uncertainty, generalisation, and learning in social interactions.

Results

Healthy participants (CON; $n=53$) and participants diagnosed with BPD ($n=50$), matched on age, gender, education, and social deprivation indices (**Table 1**), were invited to participate in a three-phase social value orientation paradigm—the Intentions Game (**Figure 1A**)—with virtual partners. In phase 1, participants made forced choices between two options for splitting points with an anonymous partner. In phase 2 participants learned to predict the decisions of a new anonymous partner using the same forced-choice setup, receiving feedback on the accuracy of their successive predictions. Notably, using a novel server architecture (Burgess et al., 2023), partners in phase 2 were configured to be approximately 50% different from the participants in terms of their choices, ensuring that all participants had to learn about their partners. Phase 3 mirrored the first, with participants informed that they were matched with a third anonymous partner, unrelated to those in phases 1 and 2. Detailed descriptions of the task can be found in the methods section and **Figure 1**. All participants also self-reported their trait paranoia, childhood trauma, trust beliefs, and trait mentalizing (see Methods).

Psychometric and Behavioural Results

Participants with BPD, compared to CON, retrospectively reported significant childhood trauma, epistemic disruptions (including mistrust and credulity), elevated referential and persecutory beliefs, and demonstrated ineffective trait mentalizing (**Table 1**). The groups did not differ in trait measures of certainty regarding self and others' mental states, nor in epistemic trust.

We analysed the 'types' of choices participants made in each phase (**Supplementary Table 1**). The interpretation of a participant's choice depends on both values in a choice. For example, a participant could make prosocial (self=5; other=5) versus individualistic (self=10; other=5) choices, or prosocial (self=10; other=10) versus competitive (self=10; other=5) choices. There were 12 of each pair in phases 1 and 3 (individualistic vs. prosocial; prosocial vs. competitive; individualistic vs. competitive).

In phase 1, both CON and BPD participants made prosocial over competitive choices with similar frequency (CON=9.67[3.62]; BPD=9.60[3.57]; $t=-0.11$, $p=0.91$). However, CON participants made significantly fewer prosocial choices when individualistic choices were available (CON=2.87[4.01]; BPD=5.22[4.54]; $t=2.75$, $p=0.007$). Both groups favoured individualistic over competitive choices with similar frequency (CON=11.03[1.95]; BPD=10.34[2.63]; $t=-1.52$, $p=0.13$). For a reaction time assessment see **Supplementary Text 1**.

Each group showed good predictive accuracy (CON=77.2%[13.9%]; BPD=72.7%[15.6%]). There was no difference in overall predictive accuracy between BPD and CON (linear estimate=2.44, 95%CI: -0.67, 5.54; $t=1.56$; $p=0.12$), nor on a trial-by-trial basis (linear estimate=0.26, 95%CI: -0.06, 0.59; $z=1.61$, $p=0.11$). All participants showed an effect of time on accuracy, such that participants became more accurate in predicting their partner over the course of phase 2 (linear estimate=0.013, 95%CI: 0.008, 0.017; $z=6.01$; $p<0.001$). Server matching between participant and

partner in phase 2 was successful, with participants being approximately 50% different to their partners with respect to the choices each would have made on each trial in phase 2 (mean similarity=0.49, SD=0.12).

In phase 3, both CON and BPD participants continued to make equally frequent prosocial versus competitive choices (CON=9.15[3.91]; BPD=9.38[3.31]; $t=-0.54$, $p=0.59$); CON participants continued to make significantly less prosocial versus individualistic choices (CON=2.03[3.45]; BPD=3.78 [4.16]; $t=2.31$, $p=0.02$). Both groups chose equally frequent individualistic versus competitive choices (CON=10.91[2.40]; BPD=10.18[2.72]; $t=-0.49$, $p=0.62$).

Computational Analysis

Over all three phases, we assumed participants and their partners used a Fehr-Schmidt utility function (Fehr & Schmidt, 1999) to calculate the utility of two options ($U_{a,\beta} = \{U_{a,\beta}^1, U_{a,\beta}^2\}$), based on the joint rewards available for both the participant, $\mathbf{R}_{ppt} = \{r_{ppt}^1, r_{ppt}^2\}$ and their partner, $\mathbf{R}_{par} = \{r_{par}^1, r_{par}^2\}$. The utility of each option was weighted based on absolute-reward gain α (how much participants care about self-earnings, irrespective of the other) and relative-reward β along a prosocial-competitive axis (how much participants care about equality of outcomes); $\beta < 0$ is prosocial, whereas $\beta > 0$ is competitive.

$$U_{\alpha,\beta} = \alpha * \mathbf{R}_{ppt} + \beta * \max(\mathbf{R}_{ppt} - \mathbf{R}_{par}, 0)$$

We then constructed four models to explain how participants used their own preferences ($\theta_{ppt}^m = \{\alpha_{ppt}^m, \beta_{ppt}^m\}$) and uncertainty over these preferences ($\theta_{ppt}^s = \{\alpha_{ppt}^s, \beta_{ppt}^s\}$) to predict and learn about the preferences of their partner (θ_{par} ; Figure 1B; see methods). Model M1 (Figure 1C) suggests that participants initially use their own preferences as a prior belief about their partner (self-insertion), which is gradually diminished during the learning process in phase 2. M1 also posits that, following learning, the inferred beliefs about a partner will influence participants' own preferences, making them more similar to their partner's preferences following observation (social contagion). According to this model, participants shift towards their partner based on their uncertainty about self and others (Figure 1E): greater uncertainty over self-preferences and increased precision in representing the other cause stronger social contagion effects.

Model M4 (Figure 1D), on the other hand, suggests that participants do not engage in these generalization processes: predictions about others are not grounded in the self, and observing others does not alter self-preferences. Models M2 and M3 allow for either self-insertion or social contagion to occur independently. Consistent with prior research, we also constructed a model that assumes the same insertion and contagion processes as M1, but along a single prosocial-competitive axis ('Beta model'; Barnby et al., 2022). The 'Beta model' is equivalent to M1 in its causal architecture (both self-insertion and social contagion are hypothesized to occur) but differs in its utility function: participants might only consider a single dimension of relative reward allocation, which is typically emphasized in previous studies (e.g., Hula et al., 2018).

All computational models were fitted and compared using a Hierarchical Bayesian Inference (HBI) algorithm which allows hierarchical parameter estimation while assuming random effects for group and individual model responsibility (Piray et al., 2019; see Methods for more information). We report individual and group-level model responsibility, in addition to exceedance probabilities between-groups to assess model dominance.

	BPD	CON	Test
DEMOGRAPHICS			
n	50	53	
Age	31.2[11.16]	30.0[8.64]	t=-0.61; p=0.54
Gender (M:F:O)	8:39:2	14:36:3	t=-1.25; p=0.21
Education (Years)	14.39(3.38)	14.8[3.10]	t=0.62; p=0.53
Soc. Dep. Index	12834.94[7911]	11967.63[7567]	t=-0.54; p=0.59
PSYCHOMETRICS			
RGPTSB	14.53[1.07]	5.33[0.72]	t=4.67; p<0.001
RGPTSA	16.79[0.96]	7.29[0.70]	t=6.47; p<0.001
CTQ (Trauma)	64.88[0.90]	42.27[0.78]	t=6.48; p<0.001
CAMSQ (Self)	4.95[1.42]	5.02[0.22]	t=-0.07; p=0.94
CAMSQ (Other)	5.32[1.42]	5.15[0.17]	t=-0.71; p=0.48
MZQ (Mentalizing)	55.94[0.59]	38.16[0.85]	t=9.39; p<0.001
ETMCQ (Mistrust)	5.25[0.93]	4.21[0.84]	t=5.27; p<0.001
ETMCQ (Trust)	4.86[1.13]	5.04[0.86]	t=-0.87; p=0.39
ETMCQ (Credulity)	4.35[1.06]	3.34[0.77]	t=4.14; p<0.001

Table 1.

Demographics of participants.

CTQ=Childhood Trauma Questionnaire, MZQ = Mentalisation Questionnaire, RGPTSB=Revised Green Paranoid Thoughts Scale (Persecutory Subscale), RGPTSA=Revised Green Paranoid Thoughts Scale (Referential Subscale), CAMSQ=Certainty About Mental States Questionnaire. ETMCQ=Epistemic Trust, Mistrust and Credulity Questionnaire, M=Male, F=Female, O=Other. For continuous variables, all means are stated with corresponding standard deviations in brackets. Significant differences are highlighted in **bold**.

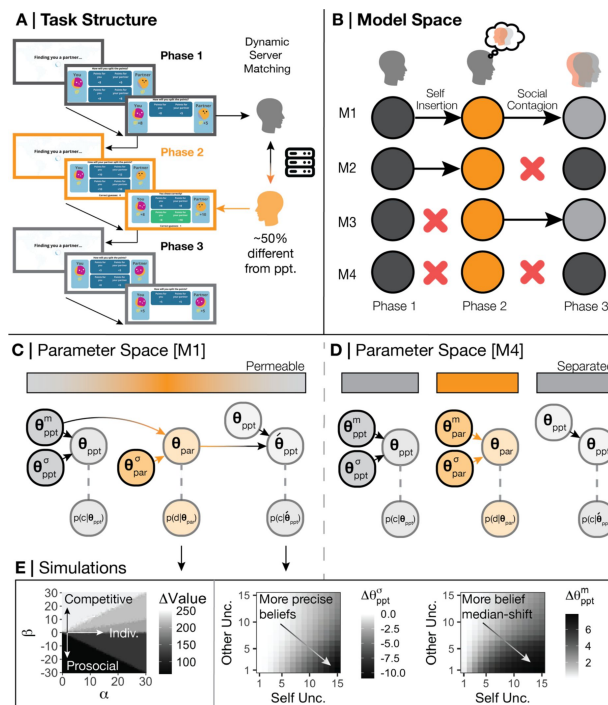


Figure 1.

Task and Model Space.

(A) Participants were invited to play a three-phase, repeated social value orientation paradigm—the Intentions Game—with virtual partners. Phase 1 of the Intentions Game lasted 36 trials and asks participants to make a forced choice between two options as to how to split points with an anonymous virtual partner. An example of a prosocial-individualistic pair of options could be (self=5, other=5) or (self=10, other=5) – if the participant chooses option 1 they could be viewed as less individualistic and more prosocial as the outcomes to the other do not change, but the self would earn less. In phase 2, lasting 54 trials, participants were asked to predict the decisions of a new anonymous partner using the same two-forced choice set-up and the same option pairs; participants were given feedback on whether they were correct or incorrect in their prediction. We used Amazon Web Services to create a novel server architecture to match participants and (virtual) partners (Burgess et al., 2023). Partners in phase 2 were matched to be approximately 50% different from the participant with respect to their choices in phase 1 to ensure all participants needed to learn about their phase 2 partner, and to provide a mechanism to examine whether beliefs about partners had an effect on the self. Phase 3 was identical to phase 1, although participants were informed that they were matched with a third anonymous partner, unconnected to the partners in phase 1 and 2. At the end of the game, if participants collected over 1000 points overall, they were entered into a lottery to win a bonus. (B) We created four models that may explain the data and to test theories of social generalization. Model M1 assumes participants are subject to both self-insertion and social-contagion, that is, participants used their own preferences as a prior about their partner in phase 2, and partner behaviour subsequently influenced participant’s preferences in phase 3. Model M4 assumes participants are subject to neither self-insertion nor social contagion, instead forming a novel prior around the phase 2 partner rather than using their own preferences and failing to be influenced by their partner after observation. Models M2 and M3 suggest participants are only explained by either self-insertion or social-contagion, not both. (C) We assume that participants choices in phase 1 are governed by both a median (θ_{ppt}^m) and standard deviation (θ_{ppt}^s). Participants insert their median preferences (θ_{ppt}^m) into their prior beliefs over their partner in phase 2, but with a different standard deviation to allow for flexibility and learning (θ_{ref}^s). The combination of the prior and posterior belief uncertainty about the partner (θ_{ref}^s ; θ_{par}^s), the precision participants have over their own preferences (θ_{ppt}^s), and the median posterior of the participant and partner (θ_{ppt}^m ; θ_{par}^m) form the new median and standard deviation over participant preferences in phase 3 (θ_{ppt}^m). (D) In contrast to M1, M4 generates a new central tendency over the partner in phase 2 (θ_{par}^m) which disconnects participant preferences and prior beliefs. M4 also assumes that the same parameters that generated participant choices in phase 1 also generate choices in phase 3. (E) Simulating our model demonstrates how different combinations of α (preferences for absolute self-reward) and β (preferences for relative reward; prosocial-competitiveness) lead to changes in the discrepancy of value between participants and partners (left panel). We also show how increasing uncertainty over self-beliefs, and higher precision over partners, causally draws participants more toward the beliefs of their partner in phase 3 and increases their precision over their phase 3 beliefs (Moutoussis et al., 2016).

	M1	M2	M3	M4	Beta	Description	Type	Phase
α_{ppt}^m	X	X	X	X		Median of absolute reward preferences	Free	1
β_{ppt}^m	X	X	X	X	X	Median of relative reward preferences	Free	1
α_{ppt}^σ	X	X	X	X		SD of absolute reward preferences	Free	1
β_{ppt}^σ	X	X	X	X	X	SD of relative reward preferences	Free	1
$\bar{\alpha}_{par}^m$			X	X		Prior beliefs median over absolute reward preferences	Free	2
$\bar{\beta}_{par}^m$			X	X		Prior beliefs median over relative reward preferences	Free	2
α_{par}^{ref}	X	X	X	X		Prior beliefs SD over absolute reward preferences	Free	2
β_{par}^{ref}	X	X	X	X	X	Prior beliefs SD over relative reward preferences	Free	2
α_{par}^m		X	X	X		Posterior belief median over absolute reward preferences	Derived	2
β_{par}^m		X	X	X	X	Posterior belief median over relative reward preferences	Derived	2
α_{par}^σ		X	X	X		Posterior belief SD over absolute reward preferences	Derived	2
β_{par}^σ		X	X	X	X	Posterior belief SD over relative reward preferences	Derived	2
$\acute{\alpha}_{ppt}^m$	X		X			Shifted median of absolute reward preferences	Derived	3
$\acute{\beta}_{ppt}^m$	X		X		X	Shifted median of relative reward preferences	Derived	3
$\acute{\alpha}_{ppt}^\sigma$	X		X			Shifted SD of absolute reward preferences	Derived	3
$\acute{\beta}_{ppt}^\sigma$	X		X		X	Shifted SD of relative reward preferences	Derived	3
No. Free	6	6	8	8	3			

Table 2.

Parameter and model specification.

Grey shading = parameters relevant to representations of the self (ppt). Orange shading = parameters relevant to representations of the other (par). Free = parameters are random variables to fit through model inversion. Derived = parameter is calculated from latent values within the model. SD = standard deviation.

Model Comparison – BPD Participants

Hold Disintegrated Self-Other Beliefs

We found that CON participants were best fit at the group level by M1 (Frequency = 0.59, Exceedance Probability = 0.98), whereas BPD participants were best fit by M4 (Frequency = 0.54, Exceedance Probability = 0.86; **Figure 2A**). We first analyse the results of these separate fits. Later, in order to assuage concerns about drawing inferences from different models, we examined the relationships between the relevant parameters when we forced all participants to be fit to each of the models (in a hierarchical manner, separated by group). In sum, our model comparison is supported by convergence in parameter values when comparisons are meaningful (see **Supplementary Materials**). We refer to both types of analysis below.

Generative Accuracy and Recovery

We simulated data for each participant using their individual parameters from the winning model within each group and refitted our models using this simulated data. Model comparison yielded very similar results (**Figure 3A**): CON synthetic participants best fit at the group level by M1 (Frequency = 0.58, Exceedance Probability = 0.98) and BPD synthetic participants best fit by M4 (Frequency = 0.57, Exceedance Probability = 0.85). The simulated data closely matched the actions of participants across all three phases (median accuracy = 0.8, SD = 0.12). In phase 2, the model-predicted total correct scores were not significantly different from observed scores (**Figure 3E**). Both model responsibility and common parameters within each dominant model were significantly associated (model confusion $p = 0.46\text{--}0.97$, $p < 0.001$; parameter recovery $p = 0.70\text{--}0.94$, $p < 0.001$; **Figure 3C**). β_{ppt}^m , preferences in phase 1 were negatively correlated with prosocial versus competitive choices ($r = -0.77$, $p < 0.001$) and individualistic versus competitive choices ($r = -0.59$, $p < 0.001$); α_{ppt}^m was positively correlated with individualistic versus competitive choices ($r = 0.53$, $p < 0.001$) and negatively correlated with prosocial versus individualistic choices ($r = -0.69$, $p < 0.001$). Given the very good to excellent performance of the models, we analysed individual parameters and simulations between groups.

Phase 1 – BPD Participants Are More Certain About Themselves

We first examined self-representations of participants in phase 1. CON participants (under model M1) and BPD participants (under M4) were equally prosocial (CON mean[β_{ppt}^m] = -7.50; BPD mean[β_{ppt}^m] = -6.59; $\Delta\mu[\beta_{ppt}^m] = 0.92$, 95%HDI: -1.24, 3.12) – both groups valued equal allocation of reward between themselves and their partners. BPD participants had lower preferences for earning higher absolute rewards (CON mean[α_{ppt}^m] = 18.41; BPD mean[α_{ppt}^m] = 10.57; $\Delta\mu[\alpha_{ppt}^m] = -7.83$, 95%HDI: -11.06, -4.75). BPD participants were also more certain about both self-preferences for absolute and relative reward ($\Delta\mu[\alpha_{ppt}^{\sigma}] = -0.89$, 95%HDI: -1.01, -0.75; $\Delta\mu[\beta_{ppt}^{\sigma}] = -0.32$, 95%HDI: -0.60, -0.04) versus CON participants (**Figure 2B**).

These differences were replicated when considering parameters between groups when we fit all participants to the same models (M1-M4; see **Table S2**).

Phase 2 – BPD Participants Use Disintegrated and Neutral Priors

We next assessed how participants generated their prior beliefs about a partner in phase 2. CON participants were best fit by M1 which assumes the same median belief participants use in phase 1 is identical to their median prior belief about their partners. In contrast, BPD participants were best fit by M4 and so generated a new median prior belief about their partners. Assessing by individual models show this was driven by expectations about a partner's prosocial-competitive preferences (relative reward; see **Table S2**).

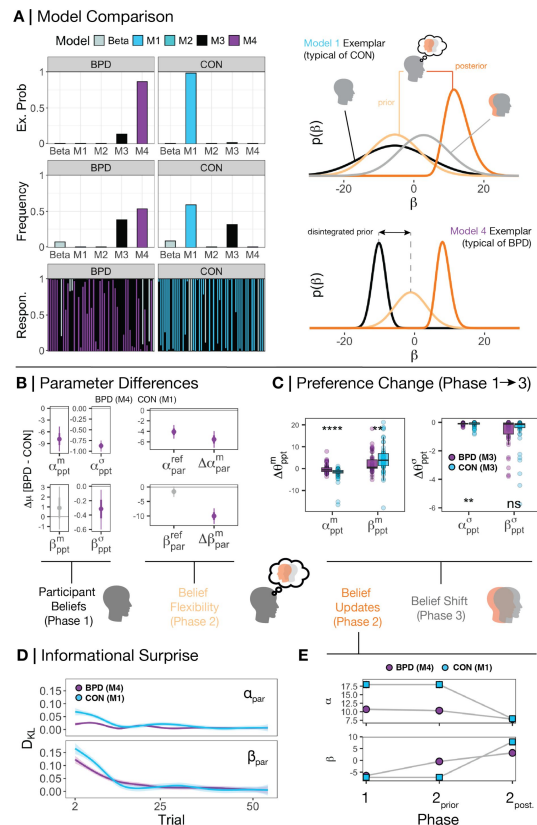


Figure 2.

Beliefs between groups and within phases.

(A) We used random-effects hierarchical model fitting and comparison to jointly estimate group level and individual level parameters based on real data from participants (Piray et al., 2021). CON participants were best fit by M1, whereas BPD participants were best fit by M4 on a group level. Looking within each model by simulating the beliefs of each participant reveals that – as expected – CON participants use the median of their self-preferences (black distribution) as a basis for their prior beliefs about partners (light orange distribution), and that the precision of their posterior beliefs about partners (dark orange distribution) and the precision of their own self preferences leads to a shifted model of the self (grey distribution). BPD participants on the other hand have a disintegrated prior over their partner which is not subject to their own self representation. Likewise, there is no change in self-preferences following learning, and thus an absence of the light grey distribution. For illustration, we focus on beliefs over relative preferences (β) and use real individual participants as exemplars for illustration. **(B)** Across models we extracted the common parameters that generate the behaviour of both CON and BPD participants – that is, their median and standard deviation over both α (absolute reward preferences) and β (relative value preferences), the flexibility over participants' prior beliefs about their partners over each dimension, and the absolute change in posterior beliefs in phase 2 over each dimension ($\Delta\alpha_{par}^m$; $\Delta\beta_{par}^m$). Using hierarchical Bayesian t-tests we demonstrated the mean difference in parameter values between groups. Purple values lower than 0 indicate that the BPD participants had significantly smaller parameter values. Here we find that BPD participants were less individualistic, equally prosocial, and more certain about their self-preferences. BPD participants were also less flexible over their beliefs about a partner's absolute reward preferences and updated their beliefs less across the board. **(C)** Examining participants under a blanket assumption that participants in both BPD and CON groups were influenced by their partner (Model M3) revealed that BPD participants were significantly less influenced by their partner across the board, both with respect to their phase 3 median and standard deviation of beliefs. Kruskal-Wallis tests were used between groups within the visualisation. $*=p<0.05$, $**=p<0.01$, $***=p<0.001$, $****=p<0.0001$. **(D)** We also calculated the Kullback-Leibler divergence (DKL) of beliefs between each trial ($t-1$ vs t) on each trial during phase 2. We observed three things: 1. All participants display larger belief updates initially, 2. all participants 'cool off' in their belief updating over the course of phase 2, and 3. BPD participants update beliefs significantly less throughout the course of phase 2 vs. CON participants. **(E)** Examining the shift in central tendencies across both groups demonstrates that between phase 1 and the start of phase 2, BPD participants shifted their central tendency only over beliefs about the relative reward preferences of partners'. They held lower expectations about a partner's absolute reward preferences consistent with their own preferences.

In BPD participants, only new beliefs about the relative reward preferences of partners differed (see **Fig 2E**) – new median priors were larger than median preferences in phase 1 ($\text{mean}[\beta_{\text{par}}^m] = -0.47$; $\Delta\mu[\beta_{\text{par}}^m - \beta_{\text{par}}^{\text{ref}}] = -6.10$, 95%HDI: -7.60, -4.60). BPD priors about their partner's relative preferences were also centred closely around 0 ($\Delta\mu[0 - \beta_{\text{par}}^m] = -0.39$, 95%HDI: -0.77, -0.05), suggesting that BPD participants entered into the interaction with very neutral priors about their partner's preferences for relative reward.

BPD participants were equally flexible around their prior beliefs about a partner's relative reward preferences ($\Delta\mu[\beta_{\text{par}}^{\text{ref}}] = -1.60$, 95%HDI: -3.42, 0.23), and were less flexible around their beliefs about a partner's absolute reward preferences ($\Delta\mu[\alpha_{\text{par}}^{\text{ref}}] = -4.09$, 95%HDI: -5.37, -2.80), versus CON (**Figure 2B**). Median belief change (from priors to posteriors) in phase 2 was lower in BPD versus CON ($\Delta\mu[\Delta\alpha_{\text{par}}^m] = -5.53$, 95%HDI: -7.20, -4.00; $\Delta\mu[\Delta\beta_{\text{par}}^m] = -10.02$, 95%HDI: -12.81, -7.30). Posterior beliefs about partner were more precise in BPD versus CON ($\Delta\mu[\alpha_{\text{par}}^{\sigma}] = -0.94$, 95%HDI: -1.50, -0.45; $\Delta\mu[\beta_{\text{par}}^{\sigma}] = -0.70$, 95%HDI: -1.20, -0.25). This is unsurprising given the disintegrated priors of the BPD group in M4, meaning they need to 'travel less' in state space. Even under assumptions of M1-M4 for both groups, BPD versus CON showed smaller posterior standard deviation in phase 2 over after learning (see **Table T2**). These results converge to suggest those with BPD form rigid posterior beliefs about prosocial-competitiveness.

We checked that these conclusions about self-insertion did not depend on the different models, we found that $\beta_{\text{par}}^{\text{ref}}$ under M1 and M2 were consistently larger in BPD versus CON. This supports the notion that new central tendencies for BPD participants in phase 2 were required for expectations about a partner's relative reward (see **Fig S10** & **Table S2**). $\alpha_{\text{par}}^{\text{ref}}$ and $\beta_{\text{par}}^{\text{ref}}$ parameters under assumptions of M1 and M2 were strongly correlated with median change in belief between phase 1 and 2 under M3 and M4, suggesting convergence in model outcome (**Fig S11**).

Analysing belief updating on a more granular trial-by-trial basis using M1 for CON and M4 for BPD revealed preference type and between-group differences in belief refinement over the course of phase 2 (**Figure 2D**). We examined this by analysing the Kullback-Leibler divergence (D_{KL}) – expected informational surprise – on each trial in Phase 2.

Across both groups and belief types informational surprise reduced over time (linear estimate $[D_{\text{KL}}] = -0.007$, 95%CI: -0.008, -0.005; $t = -7.60$, $p < 0.001$). Beliefs about a partner's relative reward preferences were updated more than absolute reward preferences (linear estimate = 0.54, 95%CI: 0.47, 0.62; $t = 14.00$, $p < 0.001$). These interacted, updating over relative vs. absolute beliefs reduced over the course of phase 2 (linear estimate = -0.013, 95%CI: -0.015, -0.011; $t = -10.81$, $p < 0.001$). These findings were supported under M1-M4 only assumptions (see **Table T3**).

BPD informational surprise is restricted over beliefs about absolute reward versus CON. CON participants remained more flexible than BPD participants along both types of preference (linear estimate $[D_{\text{KL}}(\alpha_{\text{par}}^m)] = 0.40$, 95%CI: 0.29, 0.51, $t = 7.18$, $p < 0.001$; linear estimate $[D_{\text{KL}}(\beta_{\text{par}}^m)] = 0.17$, 95%CI: 0.29, 0.51, $t = 3.06$, $p = 0.002$). There was a group by time interaction, such that the difference between groups, and the magnitude of belief updates decreased over time (linear estimate $[D_{\text{KL}}(\alpha_{\text{par}}^m)] = -0.009$, 95%CI: -0.012, -0.006, $t = -5.30$, $p < 0.001$; linear estimate $[D_{\text{KL}}(\beta_{\text{par}}^m)] = -0.004$, 95%CI: -0.008, -0.001, $t = -2.78$, $p = 0.005$) – CON participants and BPD participants eventually converged to an equivalent updating schedule (**Figure 2D**).

Assessing this same relationship under M1- and M2-only assumptions reveals a replication of this group effect for absolute reward, but the effect is reversed for relative reward (see **Table S3**). This accords with the context of each model, where under M1 and M2, BPD participants had larger phase 2 prior flexibility over relative reward (leading to larger initial surprise), which was better accounted for by a new central tendency under M4 during model comparison. When comparing

both groups under M1-M4 informational surprise over absolute reward was consistently restricted in BPD (**Table S3** [↗](#)), suggesting a diminished weight of this preference when forming beliefs about an other.

We explored how beliefs and choices were associated with reaction times, showing that belief updates and reaction times were coupled over the course of phase 2 and related to participant-partner similarity (**Figure S9** [↗](#)).

Phase 3 – BPD Participants Are Less Influenced by Partners

In the dominant model for the BPD group—M4—participants are not influenced in their phase 3 choices following exposure to their partner in phase 2. To further confirm this we also analysed absolute change in median participant beliefs between phase 1 and 3 under the assumption that M1 and M3 was the dominant model for both groups (that allow for contagion to occur). This analysis aligns with our primary model comparison using M1 for CON and M4 for BPD (**Figure 2C** [↗](#)). CON participants altered their median beliefs between phase 1 and 3 more than BPD participants (M1: linear estimate = 0.67, 95%CI: 0.16, 1.19; $t = 2.57$, $p = 0.011$; M3: linear estimate = 1.75, 95%CI: 0.73, 2.79; $t = 3.36$, $p < 0.001$). Relative reward was overall more susceptible to contagion versus absolute reward (M1: linear estimate = 1.40, 95%CI: 0.88, 1.92; $t = 5.34$, $p < 0.001$; M3: linear estimate = 2.60, 95%CI: 1.57, 3.63; $t = 4.98$, $p < 0.001$). There was an interaction between group and belief type under M3 (M3: linear estimate = 2.13, 95%CI: 0.09, 4.18, $t = 2.06$, $p = 0.041$) but not M1. There was a main effect of belief type on precision under M3 (linear estimate = 0.47, 95%CI: 0.07, 0.87, $t = 2.34$, $p = 0.02$) but not M1; relative reward preferences became more precise across the board. Derived model estimates of preference change between phase 1 and 3 strongly correlated between M1 and M3 along both belief types (see **Table S2** [↗](#) and **Fig S11** [↗](#)).

Exploratory Parameter Associations

We explored whether social contagion may be restricted as a result of childhood trauma, paranoia, and less effective trait mentalizing. We collected psychometric data from participants prior to entering the task and asked participants to attribute explicit intentions to their partner after phase 2. For raw parameter correlations with all psychometric measures see **Figure S7**. [↗](#) All analyses were corrected for False Discovery Rate (FDR; $p[\text{fdr}]$).

We explored psychometric associations with social contagion under the assumption of M3 for all participants (where everyone is able to be influenced by their partner). We conducted partial correlation analyses to estimate relationships conditional on all other associations and retained all that survived bootstrapping (1000 reps), permutation testing (1000 reps), and subsequent FDR correction. Persecution and CTQ scores were both moderately associated with MZQ scores (RGPTSB $r = 0.41$, 95%CI: 0.23, 0.60, $p = 0.004$, $p[\text{fdr}] = 0.043$; CTQ $r = 0.354$ 95%CI: 0.13, 0.56, $p = 0.019$, $p[\text{fdr}] = 0.02$). MZQ scores were in turn moderately and negatively associated with shifts in prosocial-competitive preferences ($\Delta \alpha_{\text{ppr}}^m$) between phase 1 and 3 ($r = -0.26$, 95%CI: -0.46 , -0.06 , $p = 0.026$, $p[\text{fdr}] = 0.043$). CTQ scores were also directly and negatively associated with shifts in individualistic preferences ($\Delta \alpha_{\text{ppr}}^m$; $r = -0.24$, 95%CI: -0.44 , -0.13 , $p = 0.052$, $p[\text{fdr}] = 0.065$). This provides some preliminary evidence that trauma impacts beliefs about individualism directly, whereas trauma and persecutory beliefs impact beliefs about prosociality through impaired trait mentalising (**Figure 4A** [↗](#)). Social contagion under M3 was highly correlated with contagion under M1 (**Fig S11** [↗](#)).

We also tested parameter influences on explicit intentional attributions in Phase 2. Attributions included the degree to which they believed their partner was motivated by harmful intent (HI) and self-interest (SI). According with prior work ([Barnby et al., 2022](#) [↗](#)), greater participant-partner disparity at the start of phase 2 was distinctly associated with HI and SI (**Figure 4B** [↗](#)), although associations did not survive FDR correction. Greater disparity of absolute preferences before

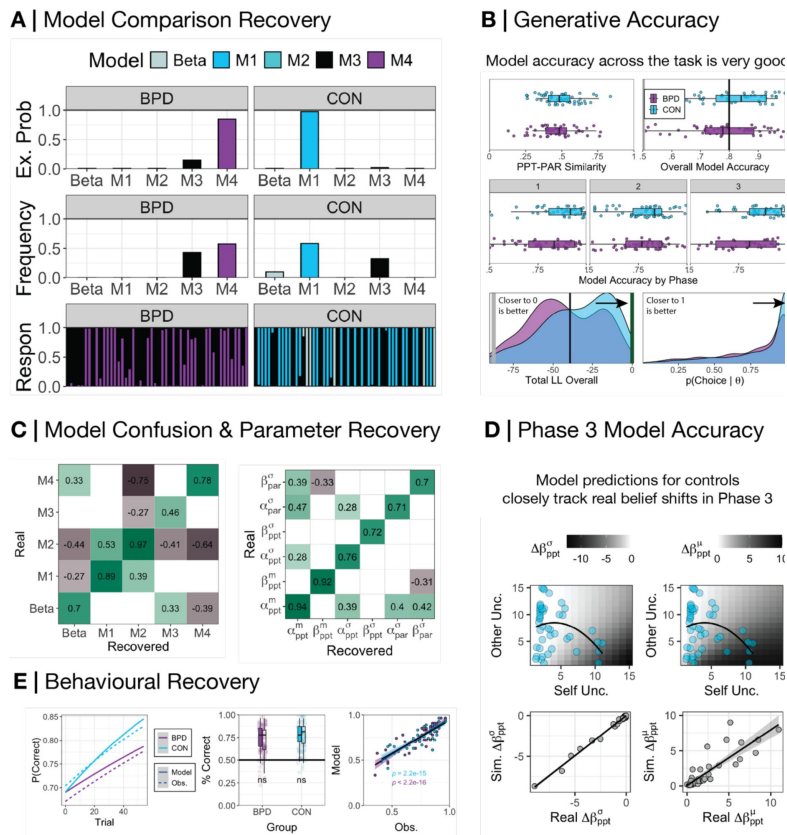


Figure 3.

Model Accuracy.

(A) We used random-effects hierarchical model fitting and comparison to jointly estimate group level and individual level parameters on simulated data (Piray et al., 2019). CON participants were best fit by M1, whereas BPD participants were best fit by M4 (B) Server matching between participant and partner in phase two was successful, with participants being approximately 50% different to their partners with respect to the choices each would have made on each trial in phase 2 (mean similarity=0.49, SD=0.12). Model accuracy across the task was very high (mean accuracy=0.8, SD=0.12). Model accuracy within each phase was very high (mean accuracy[phase1]=0.83, SD[phase1]=0.16; mean accuracy[phase2]=0.77, SD[phase2]=0.14; mean accuracy[phase3]=0.82, SD[phase3]=0.17). Loglikelihood values were also well above what would be expected had the model fitted the data by chance (median=-40.68, SD=22.7; chance value=-87.33). Choice probabilities generated by the model on each trial were also well above chance thresholds (median=0.91, SD=0.24; chance value=0.5). (C) The spearman association between the responsibility allocated for each participant during real and recovered model comparison was highly correlated on the diagonal. There was some correlation between M1-M2 but this was due to M2 being a nested model of M1, sharing similar free parameters; this was not worrying in light of excellent model identifiability overall in the synthetic comparison. Associations between real and recovered parameters from the dominant model within each BPD and CON participants was very high with few cross correlations on the off-diagonal. In both confusion and parameter recovery matrices, white spaces indicate insignificant associations at the $p > 0.01$ level. (D) (top panel) The relationship between uncertainty over the self and uncertainty over the other with respect to the change in the precision (left) and median-shift (right) in phase 3 relative reward preferences ($\hat{\beta}_{ppt}$). CON participant self and other uncertainty is overlaid onto the plot to demonstrate the degree to which their beliefs *should* change in phase 3 according to the model. (bottom panel) Correlating the model-predicted median shift in beliefs and derived change in beliefs between phase 1 and 3 demonstrates a very strong association ($r = 0.88$, $p < 0.001$). For the purposes of visualisation we cap real and simulated values <15 for compactness, although the true correlation reported is irrelevant to this visual constraint. (E) (left panel) We overlay model-predicted (solid line) and real observed (dashed line) trial-by-trial probabilities extracted from a linear model for a correct prediction by participants. For raw trial by trial updating see [Supplementary Figure 5](#). Both closely match. (middle panel) There was no significant difference (ns) for BPD and CON participants with respect to their total correct answers over phase 2. (right panel) Model-predicted and real observations in phase 2 total scores were highly correlated in both groups (CON $r=0.84$, $p<0.001$; BPD $r=0.89$, $p<0.001$).

learning was associated with reduced attributions of SI ($\rho[|\alpha_{par}^m - \alpha_{ppt}^m|] = -0.23, p=0.02, p[fdr]=0.08$), and greater disparity of relative preferences before learning exaggerated attributions of HI ($\rho[|\beta_{par}^m - \beta_{ppt}^m|] = 0.21, p=0.04, p[fdr]=0.08$). This is likely due to partners being significantly less individualistic and prosocial on average compared to participants ($\mu[\alpha] = -5.50, 95\%HDI: -7.60, -3.60; \Delta\mu[\beta] = 12, 95\%HDI: 9.70, 14.00$); partners are correctly recognised as less selfish and more competitive.

Greater prior uncertainty (before interaction) over a partner's relative preferences was associated with increased HI ($\rho[\beta_{par}^{ref}] = 0.25, p=0.01, p[fdr]=0.04$) but not SI, according with prior work (Barnby et al., 2022 [↗](#)). There was no uncorrected association between prior uncertainty over absolute preferences with either attribution. This suggests that expectations of greater difference, irrespective of one's true difference between self-other, may exaggerate beliefs about the intentional harm of others.

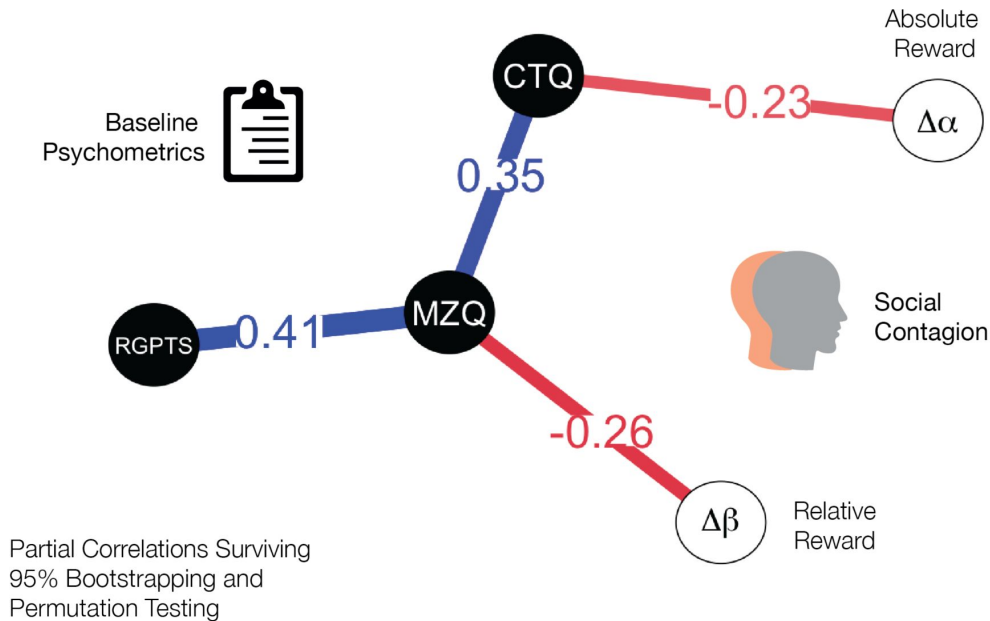
Discussion

We built and tested a theory of interpersonal generalisation in a population of matched participants with (BPD) and without (CON) a diagnosis of borderline personality disorder using the Intentions Game, a three-phase social value orientation task. Both groups demonstrated equivalent behavioural accuracy but employed different strategies. CON participants used a process of self-other generalization to predict and align with their partners, while BPD participants maintained distinct representations of self and other, particularly over joint reward outcomes. As a whole, all participants were more sensitive to updates about joint versus absolute outcomes, with BPD participants particularly concerned with how outcomes relatively effected self and other. Our exploratory findings also indicate that retrospectively reported childhood trauma and persecutory beliefs were linked to reduced trait mentalising, which was subsequently associated with diminished self-change. Collectively, our results integrate prior findings in BPD and provide a formal account of social information generalisation in humans, alongside a concise social paradigm to test these processes.

The data replicate models of social generalisation that have focused on individual processes of self-insertion and contagion, extending these theories by demonstrating both processes in conjunction. Models of self-insertion directly map participant preferences onto prior beliefs about others, which has been used to explain increased reaction times in observational learning of others' snack food preferences (Tarantola et al., 2017 [↗](#)), as well as improved predictive accuracy when matched with individuals of similar social values (Barnby et al., 2022 [↗](#)). Both findings are replicated in this study. Although we did not explicitly model reaction times, we observed an interaction between reaction time reductions over time and interpersonal similarity at baseline. In tandem, computational models of social contagion have focused on intertemporal discounting (Moutoussis et al., 2016 [↗](#)) – with behavioural studies also focusing on effort-based reward (Devaine & Daunizeu, 2017) and moral preferences (Yu et al., 2021 [↗](#)) – and explain shifts in self-preferences as a function of uncertainty regarding self and others. In both the dominant (M1) and sub-dominant (M3) models that best explained data in healthy participants, shifts in self-beliefs were also influenced by representational uncertainty of self and other: greater self-uncertainty and reduced other uncertainty led to larger shifts in social preferences.

The data also align with prior research on social impression formation, which suggests that humans form rapid evaluations of others that are refined over time (Bone et al., 2021 [↗](#); Moutoussis et al., 2023 [↗](#)). This initial 'heating' and subsequent 'cooling' of beliefs corresponds to the computational complexity employed: model-based strategies are typically used early in interactions, transitioning to simpler, model-free computations once a partner's behaviour

A | Social Contagion Modulation Under M3



B | Phase 2 Attributional Change

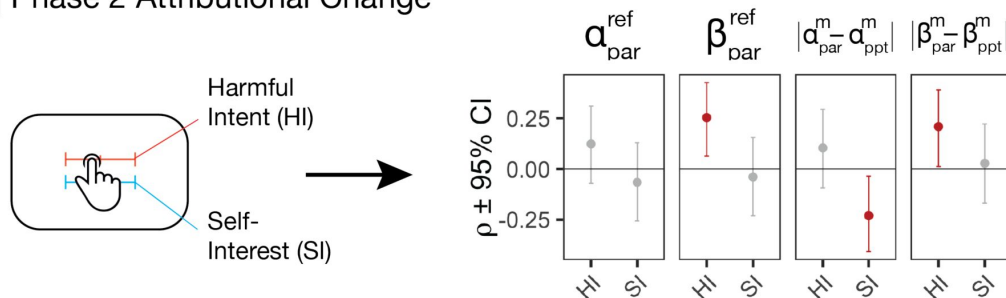


Figure 4.

Psychometric correlations.

(A) We conducted partial correlations between MZQ, CTQ, RGPTSB and changes in α_{ppt}^m ; β_{ppt}^m between phase 1 and 3 across the entire population under M3 to allow parameter comparison on the same scale. We performed bootstrapped resampling (1000 reps) and kept the edges that survived for visualisation. We then performed permutation analysis (1000 reps) to control for multiple comparisons within the partial network. Blue edges = positive partial correlation; Red edges = negative partial correlation. Note: the edge between the CTQ and $\Delta \alpha_{ppt}^m$ missed conventional significance after correction ($p=0.065$) (B). Uncorrected ranked spearman correlations between belief flexibility ($|\alpha_{par}^{ref}; \beta_{par}^{ref}|$) and absolute partner-participant dissimilarity ($|\alpha_{par}^m - \alpha_{ppt}^m|$; $|\beta_{par}^m - \beta_{ppt}^m|$) – with respect to server-derived parameters – in phase 2, controlling for group status. Only flexibility over relative reward preferences in phase 2 was associated with harmful intent attributions. Increased absolute participant-partner dissimilarity was associated with lower self-interest attributions, and increased relative participant-partner dissimilarity was associated with high harmful intent attributions.

becomes predictable (Gęsiarz & Crockett, 2015 [↗](#); Guennouni & Speekenbrink, 2022 [↗](#)). Our findings support this framework, demonstrating initial variability early in interactions followed by steady updating.

Disruptions in self-to-other generalization provide an explanation for previous computational findings related to task-based mentalizing in BPD. Studies tracking observational mentalizing reveal that individuals with BPD, compared to those without, place greater emphasis on social over internal reward cues when learning (Henco et al., 2020 [↗](#); Fineberg et al., 2018 [↗](#)). Those with BPD have been shown to exhibit reduced belief adaptation (Siegel et al., 2020 [↗](#)) along with ‘splitting’ of latent social representations (Story et al., 2024a). BPD is also shown to be associated with overgeneralisation in self-to-other belief updates about individual outcomes when using a one-sided reward structure (where participant responses had no bearing on outcomes for the partner; Story et al., 2024b). Our analyses show that those with BPD are equal to controls in their generalisation of absolute reward (outcomes that only affect one player) but disintegrate beliefs about relative reward (outcomes that affect both players) through adoption of a new, neutral belief. We interpret this together in two ways: 1. There is a strong concern about social relativity when those with BPD form beliefs about others, 2. The absence of self-insertion when predicting relative outcomes may predispose to brittle or ‘split’ beliefs. In other words, those with BPD assume ambiguity about the social relativity preferences of another (i.e. how prosocial or punitive) and are quicker to settle on an explanation to resolve this. Although self-insertion may be counter-intuitive to rational belief formation, it has important implications for sustaining adaptive, trusting social bonds via information moderation.

Those with a diagnosis of BPD also show reduced permeability in other-to-self generalising. While prior research has predominantly focused on how those with BPD use information to form impressions, it has not typically been examined whether these impressions affect the self. In interactive trust paradigms, neural responses to monetary offers from others to the self were substantially blunted in individuals with BPD compared to those without (King-Casas et al., 2008 [↗](#)). Similarly, in non-social reward tasks, those with BPD show reduced neural feedback-related negativity amplitudes, which obstructs feedback-related self-change (Stewart et al., 2019 [↗](#); Vega et al., 2013 [↗](#)). Our results suggest a mechanistic basis for social contagion, indicating that self-rigidity prevents observed social behaviours from generalizing to the self, potentially exacerbated by childhood trauma, paranoia, and impaired mentalizing capabilities. Resistance to social influence may serve as a protective response but can also contribute to the pervasive loneliness experienced by individuals with BPD, even in the absence of social isolation (Liebke et al., 2017 [↗](#)).

Notably, despite differing strategies, those with BPD achieved similar accuracy to CON participant. While all participants were more concerned with relative vs. absolute reward, those with BPD changed their strategy contingent on this focus. Practically this difference in BPD is captured either through disintegrated priors with a new median (M4) or very noisy, but integrated priors over partners (M1) if we assume M1 can account for the full population. In either case, the algorithm underlying the computational goal for BPD participants is far higher in entropy and emphasises a less stable or reliable process of inference. It is important to assess this mechanism alongside momentary assessments of mood to understand whether more entropic learning processes contribute to distressing mood fluctuation.

Clinical implications of our work underscore the importance of consistency and stability in clinical support for individuals with a diagnosis of BPD. Encouragingly, we found that those with BPD were not entirely impermeable to observed behaviour, suggesting that consistent external models of trust could be internalized over time. Restoring a stable sense of self through social learning and effective mentalizing (Nolte et al., 2023 [↗](#)), along with a consistent focus on differentiating self from other (de Meulemeester et al., 2021), are central to mentalization-based therapies (Bateman & Fonagy, 2010 [↗](#); Smits et al., 2024) and other evidence-based treatments for BPD. We hope that our

paradigm and model can offer insights into the effectiveness of these and other therapies in driving mechanistic psychological change. A key task for future work will be to assess whether generalisation principles may be restored in within-individuals with a diagnosis of BPD following intervention.

More broadly, our model bridges formal theories of associative learning and social cognition. Reinforcement learning approaches have effectively organized theories around uncertainty navigation in non-social contexts (Piray et al., 2021 [↗](#); Zika, 2023 [↗](#)). However, humans do not function in isolation. Bayesian models of internal and external social beliefs are better suited to capture the dynamic nature of time, context, and uncertainty during interactions (FeldmanHall & Nassar, 2021 [↗](#); Velez & Gweon, 2021), where joint reward rather than individual reward may be particularly salient (Barnby et al., 2023 [↗](#)). Our paradigm is concise, visually engaging, includes straightforward rules and instructions, and allows for tight experimental control over partner similarity. Our model and paradigm effectively capture core social psychological principles grounded in general computational approaches to learning and uncertainty, elucidating key aspects of human social interaction and exchange.

We note some limitations to our study. Primarily, we focused on the ability of individuals to integrate their self-concept into beliefs about others. It is also possible that humans possess strong, salient representations of others (or groups of others) that serve as dominant templates for learning. This may be particularly relevant for individuals with BPD, who will often have interpersonal experiences of abuse, neglect, or other forms of distress. The use of a salient, negative other-prior as a basis for learning was not measured in this study, but it may explain the ambivalent prior observed in phase 2, where a mixture of self and notional other influences belief formation, leading to rigid belief updating. Individuals with BPD may integrate priors from different sources as a mixture. We can simulate this by modelling a causal framework that incorporates priors based on both self and a strong memory impression of a notional other (**Figure S3** [↗](#)). However, a strength of our data is that we observed impression formation independent of valence—impressions were formed regardless of whether a partner was more or less prosocial or selfish than the participant (**Figure S4** [↗](#)). This supports our hypothesis that a vulnerable self-model and lack of self-insertion contribute to the formation of overly precise beliefs during learning as a means of rapidly reducing uncertainty. Even if a mixture model better explains the ambivalent prior in phase 2, it would still support a general hypothesis about the fractured concept of self and other in BPD.

Another strength of our work is demonstrating processes of self-insertion and contagion under minimal interaction conditions: simple observation alone was sufficient to elicit both processes. However, this is also a limitation. While we predict that these processes will apply in more naturalistic settings, this has yet to be tested, and it remains unclear whether these effects will persist in richer conditions, particularly when higher affective arousal and challenges to mentalising are present. Lastly, the action space and parameters governing choice in our study were quite simple—two actions influenced by two parameters. This was a deliberate computational choice to avoid overly complex action spaces that may be difficult to fit to real human data, and which might fail to capture how these mechanisms operate in the context of increasing action and model complexity. As a whole, our findings open new possibilities for testing how social uncertainty across the lifespan (e.g. in adolescence; Sebastian et al., 2008 [↗](#)), and in the context of ill-health, may explain the formation and maintenance of healthy social bonds as well as their disruption.

Finally, a limitation may be that behaviour in tasks based on economic preferences may not have clinical validity. This issue is central to the field of computational psychiatry, much of which is based on generalising from tasks like that within this paper and discussing correlations with psychometric measures. Extrapolating economic tasks into the real world has been the topic of discussion for the many reviews on computational psychiatry (e.g. Montague et al., 2012;

Hitchcock et al., 2022; Huys et al., 2016). We note a strength of this work is the use of model comparison to understand causal algorithmic differences between those with BPD and matched healthy controls. Nevertheless, we wish to further pursue how latent characteristics captured in our models may directly relate to real-world affective change.

Materials and Methods

Participants

We used a case-control, between-subjects design with 103 participants: a control group from the general population (N = 53) and a clinical group diagnosed with BPD (N = 50). Both groups were recruited for a larger study investigating social exchanges in BPD and Anti-Social Personality Disorder (approved by the Research Ethics Committee for Wales, 12/WA/0283). The control and clinical groups were matched on age, sex, years in education, and the English Indices of Deprivation based on the 2019 census (IoD 2019; Ministry of Housing, Communities & Local Government, 2019). Participants received £70 compensation for completing questionnaires and online tasks which included the Intentions Game. They also received a performance bonus if they were entered into the lottery for surpassing 1000 points over the course of the game.

Participants for the control group were recruited through an advertisement on the Call For Participants website (<https://www.callforparticipants.com>), local community services and adult schools. Inclusion criteria required control participants to have no pre-existing or current diagnoses of mental health disorders, neurological disorders, or traumatic brain injuries. Additionally, control participants must not have been currently in therapy or taking medication for any psychiatric disorders.

The majority of BPD participants were recruited through referrals by psychiatrists, psychotherapists, and trainee clinical psychologists within personality disorder services across 9 NHS Foundation Trusts in the London, and 3 NHS Foundation Trusts across England (Devon, Merseyside, Cambridgeshire). Four BPD participants were also recruited by self-referral through the UCLH website, where the study was advertised. To be included in the study, all participants needed to have, or meet criteria for, a primary diagnosis of BPD (or emotionally-unstable personality disorder or complex emotional needs) based on a professional clinical assessment conducted by the referring NHS trust (for self-referrals, the presence of a recent diagnosis was ascertained through thorough discussion with the participant, whereby two of the four also provided clinical notes). The patient participants also had to be under the care of the referring trust or have a general practitioner whose details they were willing to provide. Individuals with psychotic or mood disorders, recent acute psychotic episodes, severe learning disability, or current or past neurological disorders were not eligible for participation and were therefore not referred by the clinical trusts.

Psychometric Measures

Green et al. Paranoid Thought Scale (GPTS)

The GPTS assesses paranoid thoughts, including ideas of social reference (scale A) and persecution (scale B), in both general and clinical populations (Green et al., 2008). Each item is scored from 0 (not at all) to 5 (totally) concerning endorsement of each item. We retained items from the GPTS that were consistent with the revised version outlined in Freeman et al., 2021 (Revised GPTS; R-GPTS). The R-GPTS has demonstrated excellent psychometric properties (Freeman et al., 2021), making it a reliable and valid tool for assessing trait paranoid thoughts in non-clinical and clinical populations.

Childhood Trauma Questionnaire (CTQ)

The Childhood Trauma Questionnaire is used to screen for maltreatment history (Bernstein et al., 2003 [↗](#)). Each item is scored from 1 (never true) to 5 (very often true). The CTQ has showed good internal consistency reliability across the five scales (Sacchi et al., 2018 [↗](#)) and good construct validity based on significant associations with stress responsivity (McMahon et al., 2022 [↗](#)), and dissociation (Nobakht et al., 2021 [↗](#)).

Certainty About Mental States Questionnaire (CAMSQ)

The CAMSQ assesses one's certainty in classifying the mental states of oneself and others at an abstract level (Müller et al., 2023 [↗](#)), e.g. 'I know what other people think of me' and 'I know my feelings'. Each subscale is scored from 1 (never) to 7 (always). In US and German samples, the CAMSQ showed high internal consistency for Self-Certainty ($\omega = .90/.88$) and Other-Certainty ($\omega = .91/.89$) subscales, and high two-week test-retest reliability for Self-Certainty ($r = .85$), Other-Certainty ($r = .78$), and Other-Self-Discrepancy ($r = .82$) scores (Müller et al., 2023 [↗](#)).

Mentalisation Questionnaire (MZQ)

The MZQ is a 15-item questionnaire assessing an individual's trait mentalizing, i.e., one's ability to understand and interpret their own and others' mental states (Hausberg et al., 2012 [↗](#)). The MZQ demonstrated good internal consistency ($\alpha = .81$) and test-retest reliability ($r = .76$), and was sensitive to change over a 6-month follow-up period and showed good criterion-related validity, distinguishing individuals with BPD from those without BPD (Hausberg et al., 2012 [↗](#)). A higher score reflects worse trait mentalizing.

Epistemic Trust, Mistrust and Credulity Questionnaire

The ETMCQ is a 15-item measure calibrated to assess trust (e.g. 'I usually ask people for advice when I have a personal problem'), mistrust (e.g. 'I'd prefer to find things out for myself on the internet rather than asking people for information'), and credulity (e.g. 'I am often considered naïve because I believe almost anything that people tell me'; Campbell et al., 2021 [↗](#)). Each item is scored from 1(Strongly Disagree) to 7(Strongly Agree).

Paradigm, procedure and server architecture

The Intentions Game is a repeated social-value orientation paradigm with three phases.

In Phase 1 of the Intentions Game, participants take on the role of the decider with an anonymous partner over 36 trials. In each trial, participants choose between two options to distribute points between themselves and their partners. Participants make 12 choices each between prosocial and competitive (e.g. Option 1=[10,10], Option 2 = [10,5]) individualistic and competitive (e.g. Option 1=[10,5], Option 2=[8,1]), and prosocial and individualistic options (e.g. Option 1=[5,5], Option 2=[10,5]). Phase 1 choices allowed experimenters to classify participants' social preferences as prosocial (preferring equal outcomes), individualistic (maximising own payoff), or competitive (maximising relative payoff difference at the cost of lower self-gain).

We included a task environment that balanced each type of choice pair (see **Supplementary Table 1** [↗](#)).

In phase 2 of the game, participants were matched with a new anonymous partner and played the role of the recipient over 54 trials. In this phase, the participants predicted which of the two options their partner would choose on each trial. Trial numerical values for self and other were identical to Phase 1. Partners' decisions were determined via a dynamic algorithm (Burgess et al., 2023 [↗](#)) to ensure partners were approximately ~50% different from the participants' based on

participants' choices in phase 1. To surmise this architecture, we implemented a version of the client-server paradigm hosted on an Amazon Web Service (AWS) LightSail server, where the webbased behavioural task (implemented with JavaScript in Gorilla.sc) acted as the client and exchanged information with a remote AWS server. The server received all anonymised behavioural data following phase 1. The Application Programming Interface (API) to interact with the server used a customizable R script (v4.3) to process the received data from the participant, and additional R scripts were used to process and generate output for the participant. A function within the backend scripts first used Bayesian inference to approximate a participant's parameters for phase 1. It then simulated what choices the participant would have made in phase 2 had the participant been in the role of the partner. The algorithm then sought to find parameters that would be at least 50% dissimilar from participant parameters with respect to the generated choices of those parameters. This allowed the task behaviour of phase 2 to be dynamically updated in response to participant choices in phase 1. This facilitated tight control over the state of the task and enabled advanced computations to be performed on participant data beyond the capabilities of a web browser.

Participants were incentivised in phase 2 to predict accurately, as accurate predictions would contribute to their total point scores (total correct answers were multiplied by 10 and added to their points) and determined their entry into the lottery to win an extra £20 Amazon voucher. After participants had made their predictions, they were given feedback informed on whether their predictions were accurate.

At the end of phase 2, participants were asked to rate (1) the extent to which they thought their partner was driven by the desire to earn points in this task overall (self-interest) and (2) the extent to which they thought their partner was driven by the desire to reduce the participant's points in this task overall (attribution of harmful intent). The answers were presented using two separate sliders from 0 to 100; the sliders were initialised to be invisible until the participants made the first click.

Phase 3 was identical to phase 1 except that participants were matched with a new anonymous partner. Participants would take on the decider role similar to phase 1 which allowed experimenters to estimate whether the observation of their partner in phase 2 had an influence on participants in phase 3.

Behavioural Analysis

All analysis was conducted in R (v. 4.3.3) on a macbook pro (M2 Max; OS=Ventura13.5). All individual numeric values extraneous of statistical tests are reported with their mean and standard deviation (mean=KCWM8V2, SD=YY). All statistical tests where dependent variables mapped one value to one participant (e.g. trait psychometric scores) were conducted as linear models, with the regression coefficient, 95% confidence interval (95%CI), t-value and p-value reported like so (linear estimate=KCWM8V2, 95%CI:AA,BB; t=CC, p=DD). When dependent variables mapped multiple values to each participant (e.g. trial-by-trial accuracy or reaction time) random-effects linear modelling was used. All correlations used Pearson estimates (r) unless distributions were non-normal, in which case Spearman-ranked correlations (r) were performed.

Model space and Computational Analysis

We apply four computational hypotheses (M1-M4) which could explain the data collected from the Intentions Game (**Figure 1** [↗](#)), centred around formal principles of self-insertion and social contagion. Self-insertion states that a self inserts their own preferences into their beliefs about others ([Anderson & Chen, 2002](#) [↗](#); [Kreuger & Clement, 1994](#) [↗](#)); Social Contagion states that a self's preferences will change when exposed to the preferences of an other ([Frith & Frith, 2012](#) [↗](#)). In each case, cognitive representations of self and other are allowed to intermingle to form a new hybrid of the two for the purposes of computational efficiency and/or social bonding.

We note some important assumptions in our notation going forward. In dyadic social interaction, both parties are trying to estimate and predict the true state (θ) of the self (θ_S) and the other (θ_O). However, this estimation is inherently imperfect. Theories of social inference need to consider three sources of noisy estimation of this quantity: the self's (s) metacognitive model of their own state, $\hat{\theta}_s$, their partner's (O) state, $\hat{\theta}_o$, and finally the experimenter's approximation of both quantities, $\hat{\theta}_{s,o}$, (Barnby et al., 2024 [\[1\]](#)). In this work we consider the experimenter's approximation of the self's state $\hat{\theta}_s$ (phase1), the self's approximation of their other $\hat{\theta}_o$ (phase 2), and how exposure to a partner may influence $\hat{\theta}_s$ (phase 3). We term the self the participant (ppt) and the other the partner (par) and assume $\theta_{ppt} = \hat{\theta}_s$ in phase 1, $\theta_{par} = \hat{\theta}_o$ in phase 2, and θ_{ppt} are the shifted participant preferences following exposure to the partner.

All models assumed a constricted Fehr-Schmidt utility function was used by participants and partners to calculate the utility of two options ($U_{\alpha,\beta} = \{U_{\alpha,\beta}^1, U_{\alpha,\beta}^2\}$) in each trial within the task.

In phase 1, participants made binary choices $c^t = \{1 \dots T\}$ about whether option 1 or option 2 should be chosen given the returns for each option pair, $\mathbf{R}^t = \{\mathbf{R}^{t,1}; \mathbf{R}^{t,2}\} = \{R_{ppt}^{t,1}, R_{par}^{t,1}, R_{ppt}^{t,2}, R_{par}^{t,2}\}$.

(1)

$$U_{\alpha,\beta}(\mathbf{R}^{t,1}) = \alpha_{ppt} * R_{ppt}^{t,1} + \beta_{ppt} * \max(R_{ppt}^{t,1} - R_{par}^{t,1}, 0)$$

$$\Delta U_{\alpha,\beta}(\mathbf{R}^t) = U_{\alpha,\beta}(\mathbf{R}^{t,1}) - U_{\alpha,\beta}(\mathbf{R}^{t,2})$$

Here, α_{ppt} describes the weight a participant places on their own payoff (in one reduced model we set $\alpha_{ppt} = 1$), and β_{ppt} the weight a participant places on their payoff relative to the payoff of their partner. Large positive or negative values of β_{ppt} indicate respectively that participants like or dislike earning more than their partner. We can therefore describe these terms α and β as reflecting preferences for absolute and relative payoffs, respectively. For efficiency we discretised states of α_{ppt} from 0.30 (increments of 0.125) and β_{ppt} from -30 to 30 (increments of 0.25).

Over this state space we can construct a belief that participants are estimated to hold which generate their choices, $\mathbf{C}$. Herein, we refer to this belief as θ_{ppt} , where θ_{ppt} is a matrix over a fixed grid of α_{ppt} and β_{ppt} values. In the models, θ_{ppt} is drawn from a normal distribution made from a central tendency, θ_{ppt}^m , and a standard deviation, θ_{ppt}^σ . The standard deviation around the central tendency allows for stochastic choice behaviour consistent with random utility models (Block, 1974 [\[2\]](#); McFadden, 1974 [\[3\]](#)). We invert the model to estimate θ_{ppt} based on a participant's choices given their likelihood of choosing $c^t = 1$:

(2)

$$p(\theta_{ppt}|\mathbf{C}) \sim N(\theta_{ppt}; \theta_{ppt}^m, \theta_{ppt}^\sigma)$$

$$p(c^t = 1|\theta_{ppt}, \mathbf{R}^t) = \sum_{\theta_{ppt}} \sigma(\Delta U_{\alpha,\beta}(\mathbf{R}^t)) \cdot \theta_{ppt}$$

$$LL = \log [p(c^t = 1|\theta_{ppt}, \mathbf{R}^t)]$$

When θ_{ppt}^{σ} is larger, a participant's choices in Phase 1 are estimated to be less deterministic and more stochastic – i.e. they are less sure about their preferences along each dimension. This consideration will become important for choices made in phase 3.

In phase 2, over 54 trials, we then model the participants binary predictions d^t , $t = \{1 \dots T\}$ about whether option 1 or 2 would be chosen by their partner given the returns $\mathbf{R}^t = \{\mathbf{R}^{t,1}; \mathbf{R}^{t,2}\} = \{R_{ppt}^{t,1}, R_{par}^{t,1}, R_{ppt}^{t,2}, R_{par}^{t,2}\}$ for each pair of options. They then were given feedback about the partner's true decision which we note as d^f . We assumed the participant predict the partner in the same way they would themselves, ranging along two dimensions, α_{par} and β_{par} which was needed to be inferred through observation, using a likelihood for d^f of $LL = \log [p(d^t = 1 | \alpha_{par}, \beta_{par}, \mathbf{R}^t)]$ using the same formula as phase 1. We note the belief about α_{par} and β_{par} together as θ_{par} , represented as a matrix over a fixed grid of α_{par} and β_{par} values.

The partner decisions, $D^t = \{d^1, d^2 \dots, d^T\}$ are then used to update the participants beliefs about the partner, written as $p(\theta_{par} | D^f)$, starting with prior $p(\theta_{par} | D^0)$. Both M1 and M2 assume participants use their own central tendency, θ_{ppt}^m , as a starting point for their prior beliefs about their partner as theoretically outlined as a self-insertion bias (Barnby et al., 2024) which draws from past computational work (Barnby et al., 2022; Tarantola et al., 2017). We also assumed participants used a new standard deviation θ_{par}^{ref} which allowed for participants to believe their partner may be different from them (belief flexibility). Therefore we have:

$$(3) \quad p(\theta_{par} | D^0) \sim N(\theta_{par}; \theta_{ppt}^m, \theta_{par}^{ref})$$

In models M3 and M4, we assume participant's may instead use a new central tendency (rather than their own) as prior beliefs over their partner. This are free parameters to be approximated, $\hat{\alpha}_{par}^m, \hat{\beta}_{par}^m$.

In all cases, we assume participants update their beliefs about their partner's social preferences given their partner's decisions D along trials 1–54 according to Bayes rule:

$$(4) \quad \theta_{par}^t = \frac{p(d^t | \theta_{par}; \mathbf{R}^t) \theta_{par}^{t-1}}{\sum_{\theta'_{par}} p(d^t | \theta'_{par}; \mathbf{R}^t) \theta_{par}^{t-1}}$$

We can then marginalise over θ_{par}^t to calculate the belief participants had over their participant's social value preferences.

We assume that participants predict the partner's decision in the next trial by calculating the probability determined by the utility differences $\Delta U_{\alpha, \beta}(\mathbf{R}^{t+1})$ as in phase 1, summed over the joint distribution of partner parameters, θ_{par}^t :

(5)

$$p(\bar{d}^{t+1} = 1 | D^t, \mathbf{R}^t) = \sum_{\theta_{par}} \sigma(\Delta U_{\alpha, \beta}(\mathbf{R}^t)) \cdot \theta_{par}^t$$

$$p(\bar{d}^{t+1} = 2 | D^t, \mathbf{R}^t) = 1 - p(\bar{d}^{t+1} = 1 | D^t, \mathbf{R}^t)$$

And then performed probability matching, so that:

(6)

$$p(\bar{d}^{t+1} = 1 | D^t, \mathbf{R}^t) = p(d^{t+1} = 1 | D^t, \mathbf{R}^t)$$

In the third phase participants are once again asked to make choices for themselves and a new anonymous partner over 36 trials with an assumed identical utility function as in phase 1. In model M1 and M3 we assume participants use a combination of their own preferences and the posterior beliefs about their partner to form a new distribution to select between the two options available on each trial. This draws from the same formulation used previously (Moutoussis et al. 2016). In essence, we state that participants know their true preferences in phase 1 but are unsure about them. The inferred partner beliefs θ_{par}^t provides information to the participant about some common preference distribution both share, which in turn informs the participant's own choices c^t , $t = \{1 \dots T\}$ in the form of an adjusted belief along each dimension for phase 3, $\hat{\alpha}_{ppt}$ and $\hat{\beta}_{ppt}$ (eq. 7), using a log likelihood of $LL = \log[p(c^t = 1 | \hat{\alpha}_{ppt}, \hat{\beta}_{ppt}, \mathbf{R}^t)]$. We refer to $\hat{\alpha}_{ppt}$ and $\hat{\beta}_{ppt}$ together as $\hat{\theta}_{ppt}$ for convenience, where $\hat{\theta}_{ppt}$ is a matrix over a grid of fixed values of $\hat{\alpha}_{ppt}$ and $\hat{\beta}_{ppt}$. To note: models M2 and M4 do not assume participants undergo this change, and instead use their original phase 1 beliefs to make choices $LL = \log[p(c^t = 1 | \alpha_{ppt}, \beta_{ppt}, \mathbf{R}^t)]$.

(7)

$$p(\hat{\theta}_{ppt} | \mathbf{C}) \sim N(\theta_{ppt}; \hat{\theta}_{ppt}^m, \hat{\theta}_{ppt}^\sigma)$$

$$\hat{\theta}_{ppt}^\sigma = (\theta_{ppt}^{\sigma^{-2}} + (2\theta_{par}^{ref^2} + \theta_{par}^{\sigma^2})^{-1})^{-1}$$

$$\hat{\theta}_{ppt}^m = \hat{\theta}_{ppt}^{\sigma^2} [(\theta_{ppt}^{\sigma^{-2}} * \theta_{ppt}^m) + (2\theta_{ref}^{\sigma^2} + \theta_{par}^{\sigma^2})^{-1} * \theta_{par}^m]$$

Where θ_{par}^{σ} and θ_{par}^m are the standard deviation and central tendency of the final posterior inference about the partner, θ_{par}^{ref} .

All computational models were fitted using a Hierarchical Bayesian Inference (HBI) algorithm which allows hierarchical parameter estimation while assuming random effects for group and individual model responsibility (Piray et al., 2019). During fitting we added a small noise floor to distributions ($2.22e^{-16}$) before normalisation for numerical stability. Parameters were estimated

using the HBI in untransformed space drawing from broad priors ($\mu_M=0$, $\sigma^2_M = 6.5$; where $M=\{M1, M2, M3, M4\}$). This process was run independently for each group. Parameters were transformed into model-relevant space for analysis. All models and hierarchical fitting were implemented in Matlab (Version R2022B). All other analyses were conducted in R (version 4.3.3; arm64 build) running on Mac OS (Ventura 13.0). We extracted individual and group level responsibility, as well as the protected exceedance probability to assess model dominance per group.

To conduct model recovery we simulated synthetic participants (CON=53; BPD=50) using their fitted parameters from the dominant model of the group (CON=M1; BPD=M4). We then performed model fitting with an identical procedure to the real behavioural data. We tested associations between model responsibility and individual parameters for the real and recovered models, as well as the association between choices and predictions made by the model from simulation and the choices and predictions made by participants in each trial.

Differences between groups for individual-level parameters were estimated using hierarchical Bayesian t-tests (Bååth, 2014 [↗](#)) and hierarchical general linear models in *rStanArm*. Differences in mean between groups ($\Delta\mu$) are additionally reported with their corresponding posterior 95% High Density Interval (95%HDI). Belief updates were calculated as the Kullback-Leibler Divergence between probabilities (P) from trial $t-1$ to t , marginalised along all possible states, $S=\{s^1, s^2, \dots, s^n\}$:

$$D_{KL}(p^t || p^{t-1}) = \sum_i p^t(s) \log \frac{p^t(s)}{p^{t-1}(s)}.$$

Open data and code:

https://github.com/josephmbarnby/SocialTransfer_Barnby_et_al_2024 [↗](#)

Acknowledgements

We would like to greatly thank all participants who took part in the research.

Additional information

CRediT

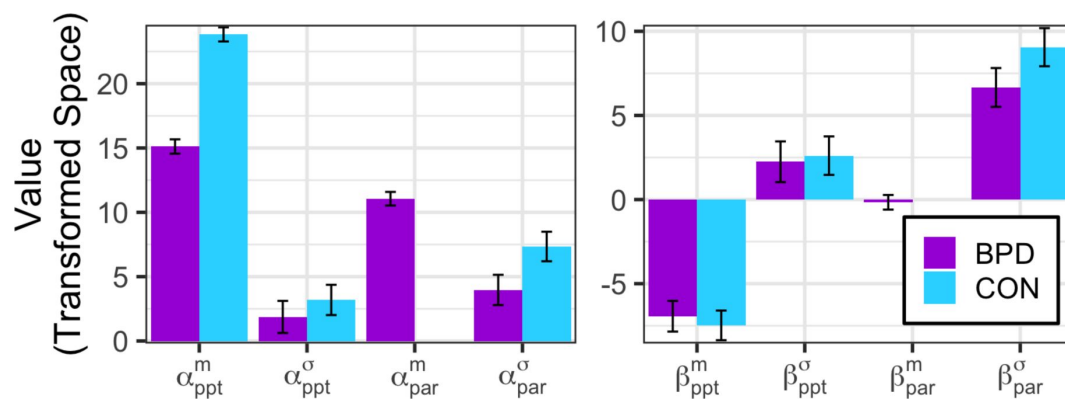
JMB: Conceptualisation, Data Curation, Investigation, Formal Analysis, Methodology, Project Administration, Software, Supervision, Visualisation, Writing – Original Draft, Writing – Review and Editing. **JN:** Investigation, Methodology, Writing – Original Draft, Writing – Review and Editing. **JG:** Conceptualisation, Investigation, Project Administration, Resources, Writing – Review and Editing. **MW:** Project Administration. **HB:** Software, Writing – Review and Editing. **LR:** Resources, Writing – Review and Editing. **GC:** Validation, Writing – Review and Editing. **JK:** Supervision, Writing – Review and Editing. **PRM:** Resources, Writing – Review and Editing. **PD:** Conceptualisation, Formal Analysis, Writing – Review and Editing. **TN:** Conceptualisation, Project Administration, Resources, Supervision, Writing – Review and Editing. **PF:** Conceptualisation, Resources, Supervision, Writing – Review and Editing.

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Project number 39072764 and of the Else Kroner Medical Scientist College “ClinbrAIn: Artificial Intelligence for Clinical Brain Research”.

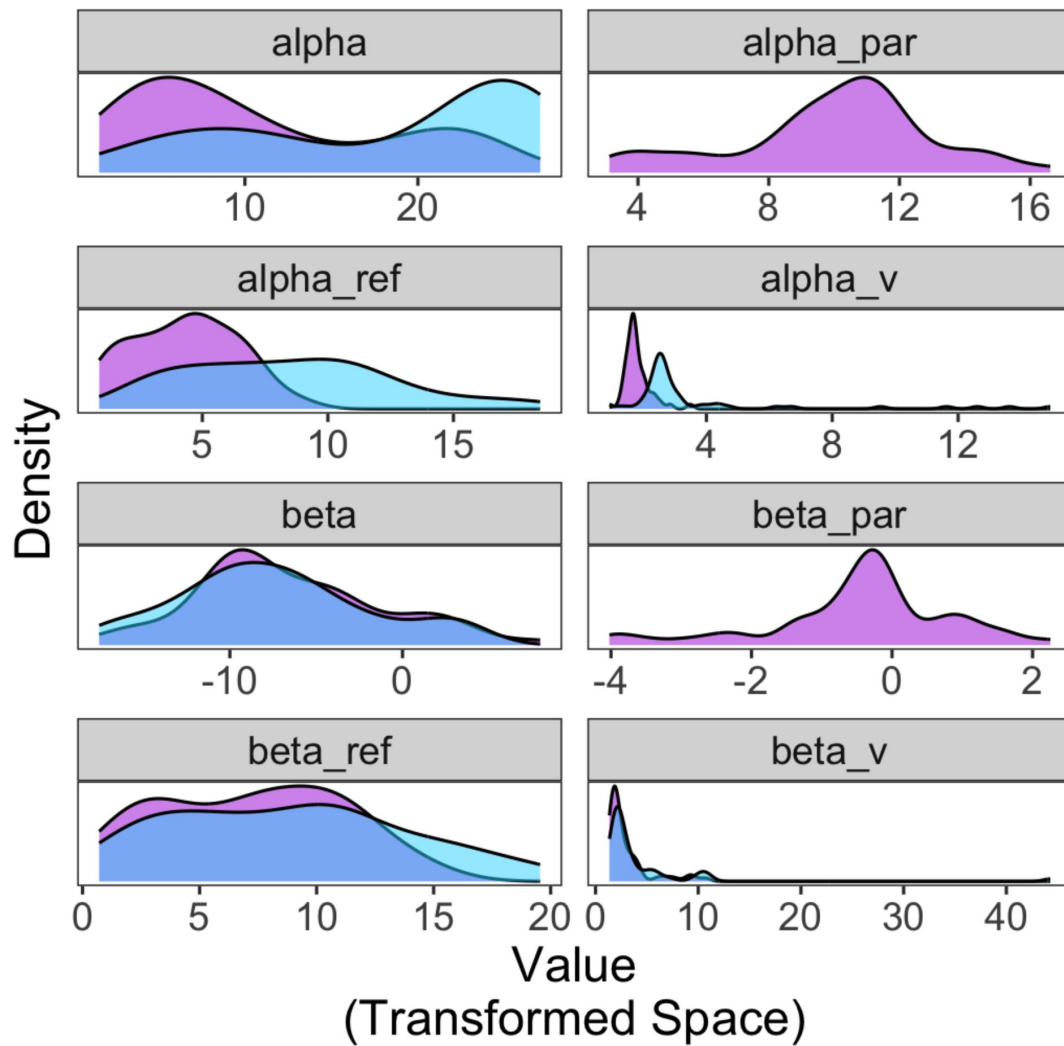
Supplementary Materials



Supplementary Figure 1.

Group Level Parameter Values.

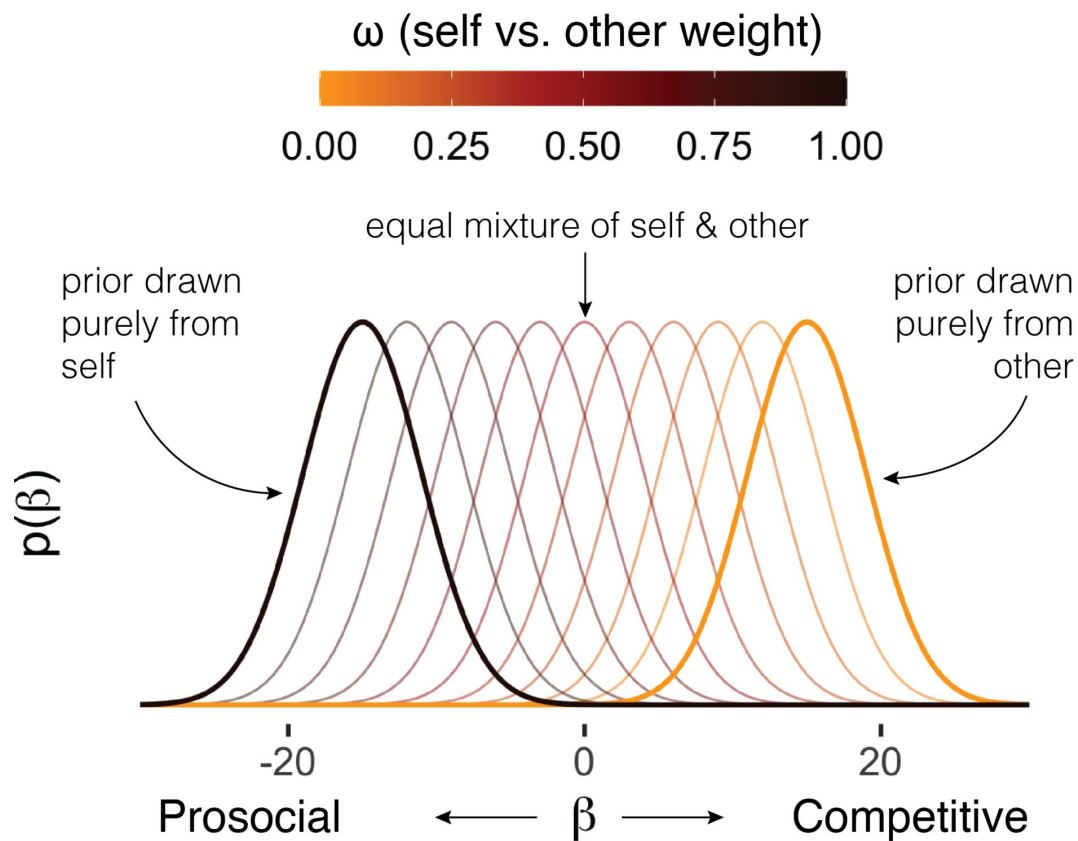
BPD participants were explained by M4 which has two extra free parameters than CON participants who were best explained by M1.



Supplementary Figure 2.

Individual Level Parameter Distributions Per Group.

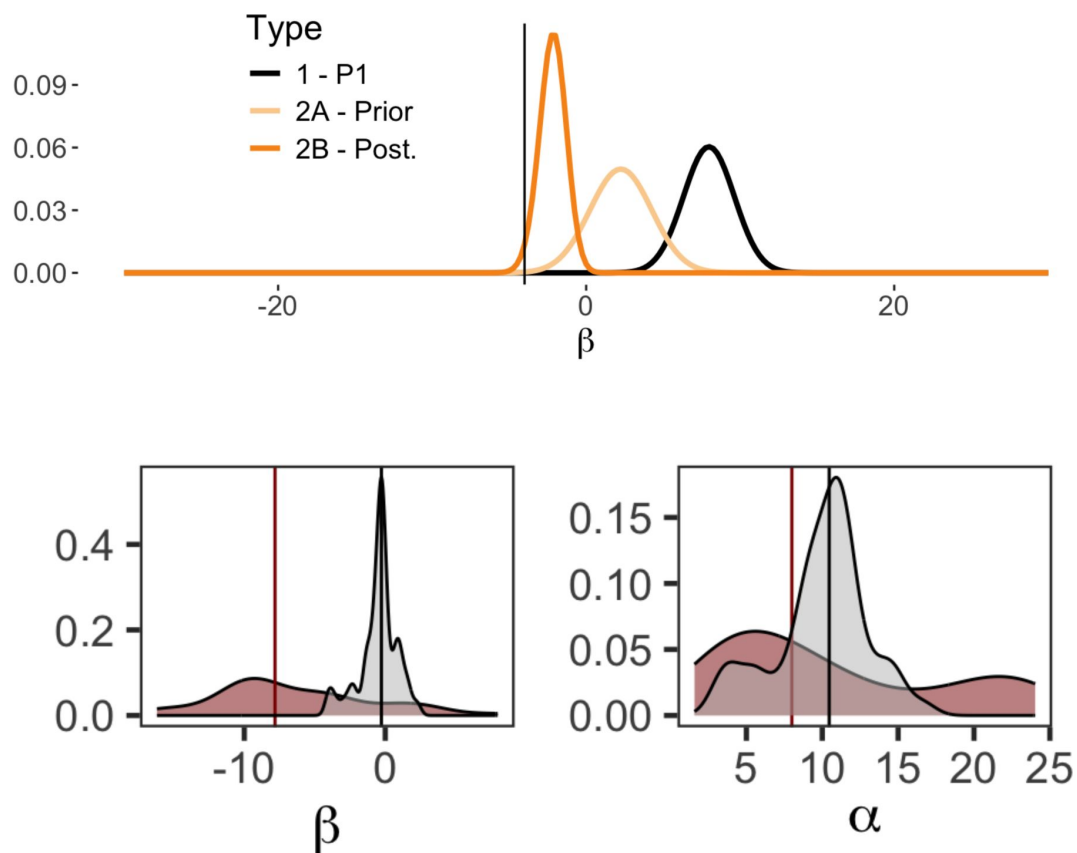
BPD (purple) participants were explained by M4 which has two extra free parameters (α_{par}) and (β_{par}) than CON participants (blue) who were best explained by M1.



Supplementary Figure 3.

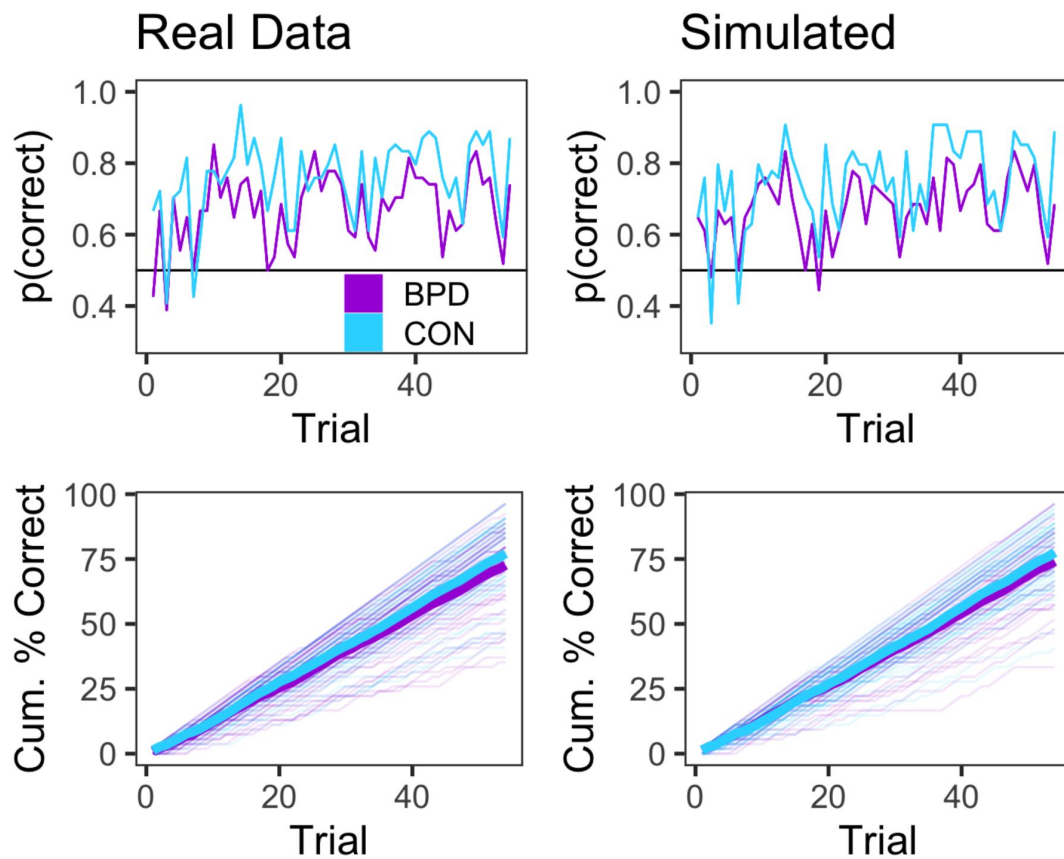
Simulation of Phase 2 priors that may be drawn from a memory of an aversive other vs from the self alone.

We can imagine a scenario where a prosocial participant (typical of BPD and CON) has a strong impression of an other from memory who is particularly aversive (competitive). Using a mixture of the median belief of the self (β_{ppt}^m ; classified in phase 1) and a mixture of the belief about how this notional competitive other (β_{par}^m) would act we can create a causal model of how priors in phase 2 about an anonymous partner might draw on different sources. Here, the median of the prior over the partner in phase 2 is a mixture of median belief of self and notional other [$\beta_{par}^m = \beta_{ppt}^m + \beta_{par}^m (1 - \omega)$]. An equal mixture of self and other belief would equally explain the naïve prior BPD participants hold over their partner in phase 2. However, as mentioned, given that BPD participants hold a naïve prior even when they are themselves competitive goes against this hypothesis. It is worth testing.



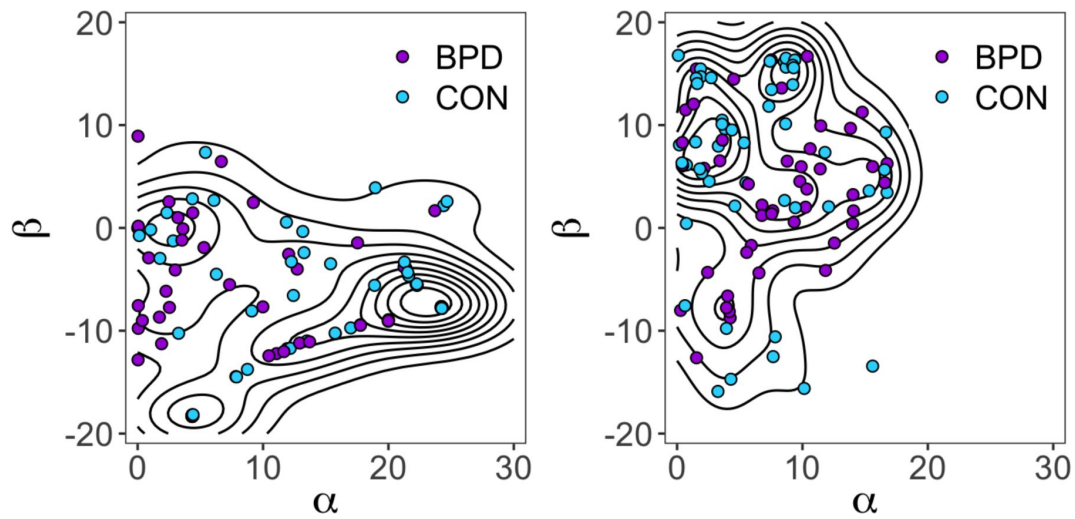
Supplementary Figure 4.

(top) Exemplar distribution from an individual with a diagnosis of BPD who was competitive in phase 1 and matched with a partner who was prosocial in phase 2. We note that irrespective of the valence of BPD participants' preferences, there was still a neutral prior generated that was not integrated into the model of self. (bottom) distribution of individual-level parameter estimates for phase 1 beliefs (self; red) and phase 2 prior beliefs (other; grey) about partners for both prosocial-competitive (left) and individualistic (right) beliefs. As reported in the main text, BPD priors about their partner's prosocial preferences were centred closely around 0 ($\Delta\mu[0 - \beta_{par}^m] = -0.39$, 95%HDI: $-0.77, -0.05$), and the shift between β_{ppt}^m and β_{par}^m is large ($\Delta\mu = 6.0$, 95%HDI: $4.7, 7.3$). This is not true of the shift between α_{ppt}^m and β_{par}^m ($\Delta\mu = -0.8$, 95%HDI: $-2.7, 1.1$).



Supplementary Figure 5.

(top panels) Raw trial-wise probability of correct responses from real and model-simulated observations for each group. Probabilities were approximated by grouping by trial across each group, summing the total correct responses and dividing by 54. (bottom panel) Cumulative percentage of correct predictions in phase 2 for each group are shown as thick solid lines. Individual cumulative scores are depicted as thin translucent lines.

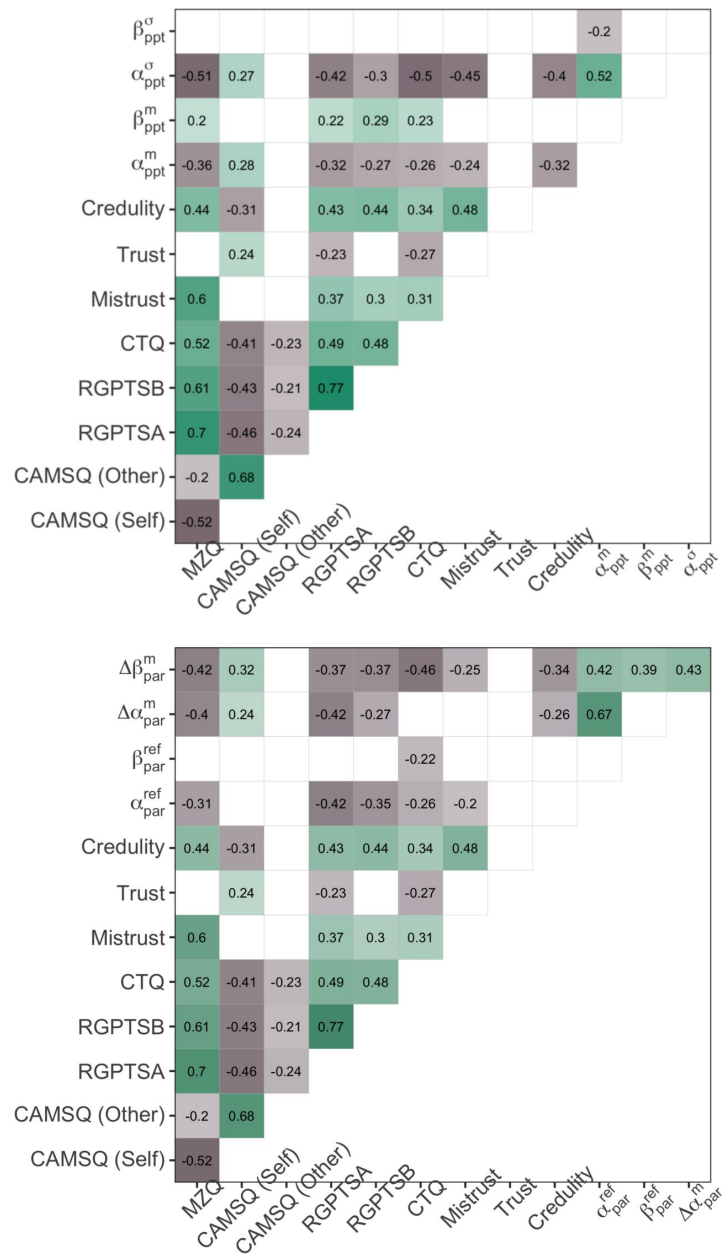


Supplementary Figure 6.

2D Distribution of participant and partner parameters estimated through Bayesian inference at the AWS server backend during the participant-partner matching protocol. As a sanity check we also assessed the degree to which server-derived participant parameters ($\hat{\alpha}_{ppt}^m, \hat{\beta}_{ppt}^m$) matched model-fitting derived model parameters; any discrepancy may

have inappropriately matched partners to participants on the server-side. We observed excellent correlations between server-derived participants (not used for analysis; only for partner matching in game) and model-derived phase 1 parameters

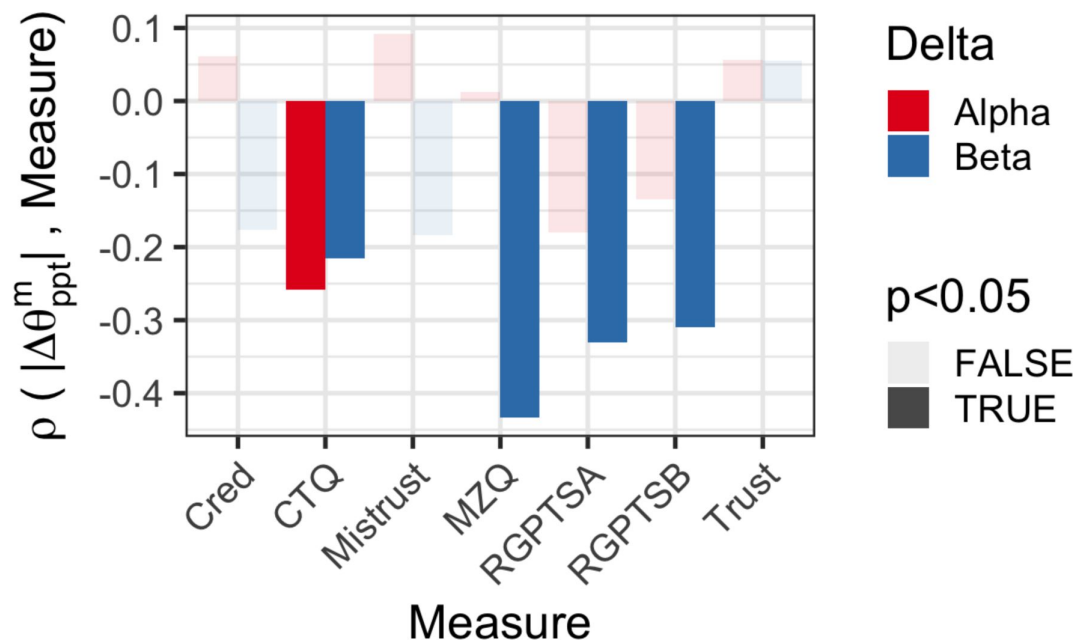
$[r(\alpha_{ppt}^m, \hat{\alpha}_{ppt}^m) = 0.85, p < 0.001; r(\beta_{ppt}^m, \hat{\beta}_{ppt}^m) = 0.83, p < 0.001]$.



Supplementary Figure 7.

Spearman Correlations Between Psychometric Scores at Baseline and Self/Other Parameters.

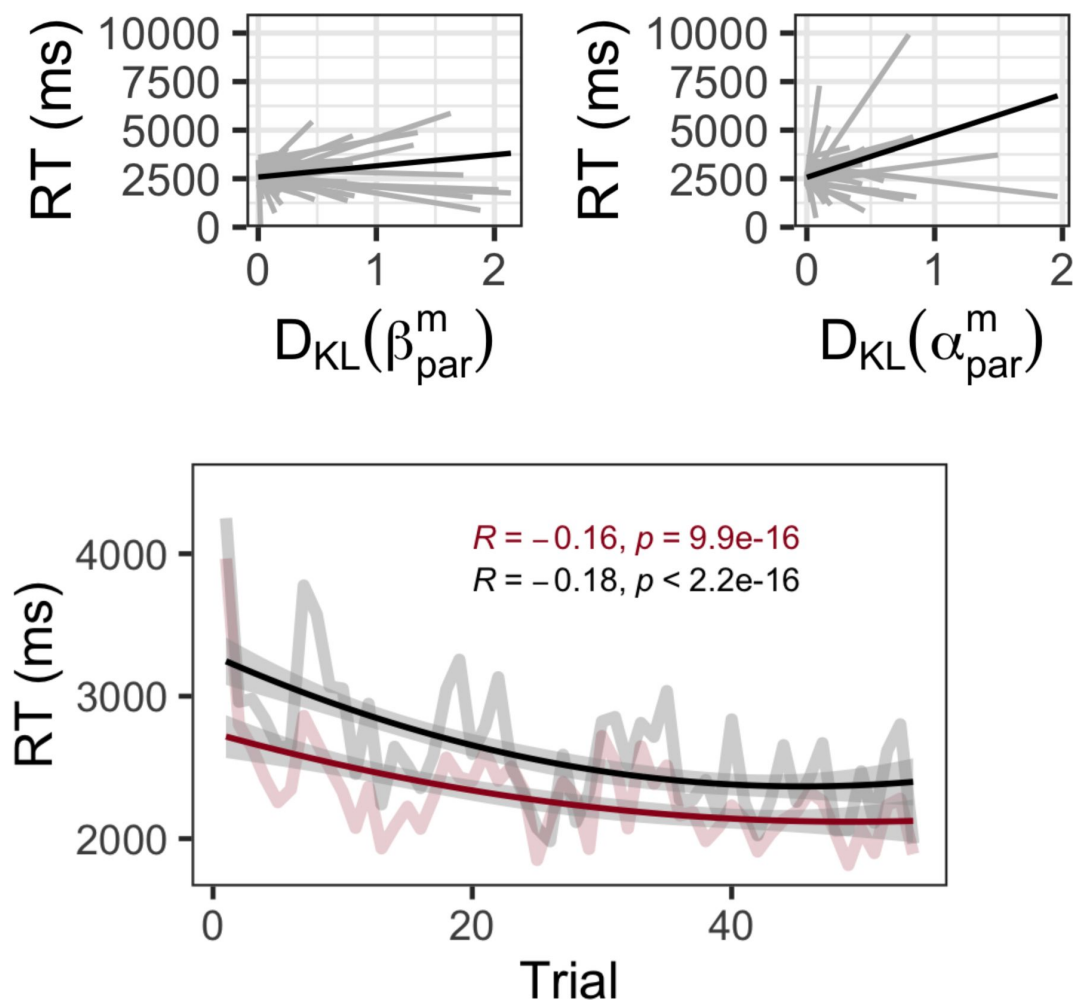
(Top) Psychometric correlations with parameters for self. **(Bottom)** Psychometric correlations with parameters for other. All correlations with p-values > 0.05 are omitted.



Supplementary Figure 8.

Uncorrected spearman's ρ between psychometric measures and change absolute change in self-preferences from phase 1 to 3.

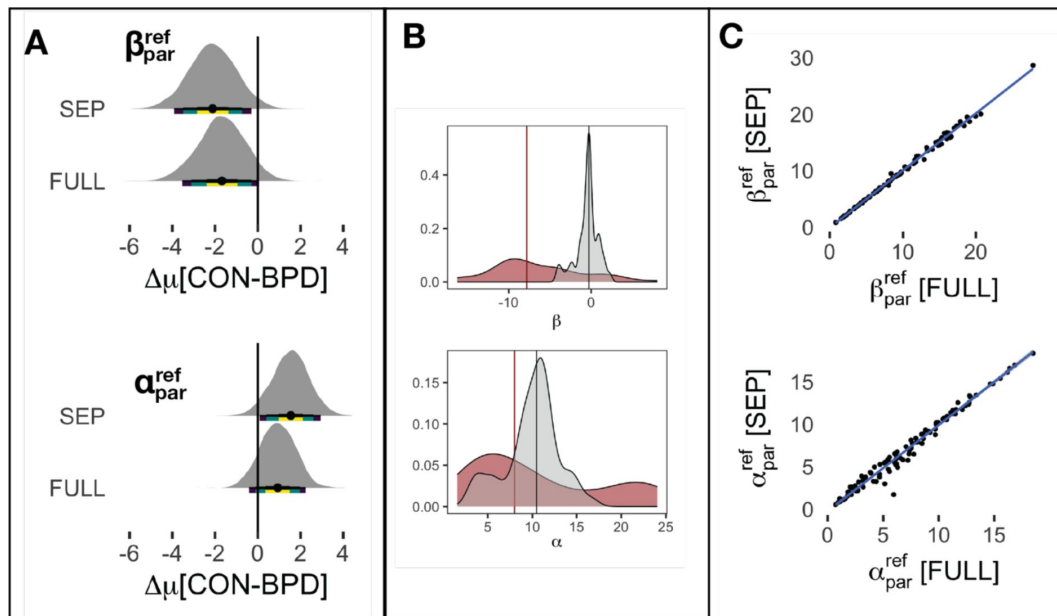
All beliefs metrics are extracted from M3 which assumes all participants engage in social contagion. Cred = Credulity. Delta = whether the shift in belief was along preferences for absolute (alpha) or relative (beta) reward.



Supplementary Figure 9.

Relationship between belief updates and reaction times.

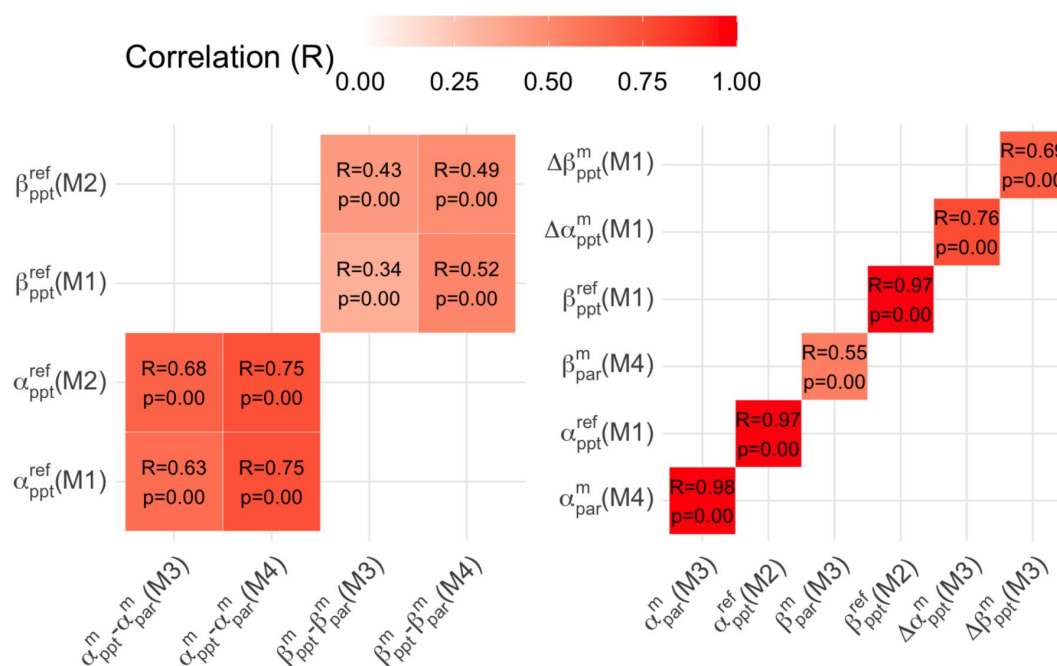
(Top) Linear random effects relationship between reaction time (ms) and belief updating. Reaction times and belief updates in phase 2 were significantly coupled, such that larger shifts in posterior beliefs along both axes were associated with larger reaction times (linear estimate [$D_{KL}(\alpha_{par}^m)$] = 0.044, 95%CI: 0.027, 0.06, $t = 5.01$, $p < 0.001$; linear estimate [$D_{KL}(\beta_{par}^m)$] = 0.021, 95%CI: 0.005, 0.039, $t = 2.49$, $p = 0.012$). Grey lines are individual participants. Black line is the average linear effect. Reaction time is capped at 10000ms for visual illustration, but linear models do not apply an upper limit. (Bottom) All participants were slower at the start of phase 2 and sped up over time (linear estimate = -15.03, 95%CI: -21.06, -8.99; $t = -4.88$, $p < 0.001$). Baseline participant-partner similarity did not have an overall effect on reaction time but did interact with trial – as participant-partner similarity (red = high similarity; black = low similarity) increased, reaction times early in phase 2 were significantly slower and this effect attenuated over time (linear estimate = -0.53, 95%CI: -0.75, -0.32; $t = -4.91$, $p < 0.001$). Reaction time did not vary between groups: both BPD and CON participants displayed the same effect.



Supplementary Figure 10.

Phase 2 prior belief flexibility following forced hierarchical fit of Model M1 to all (FULL) participants and separate (SEP) groups.

(A) Bayesian general linear model estimates of the differences in the mean of β_{par}^{ref} (top) and α_{par}^{ref} (bottom) between CON and BPD participants for both full (β_{par}^{ref} : coefficient = -1.7, 90%HDI = -3.1, -0.3; α_{par}^{ref} : coefficient = 0.9, 90%HDI = -0.1, 2.0) and separate (β_{par}^{ref} : coefficient = -2.1, 90%HDI = -3.5, -0.7; α_{par}^{ref} : coefficient = 1.5, 90%HDI = 0.4, 2.6) hierarchical fits. β_{par}^{ref} was consistently larger in BPD participants across both SEP and FULL fits. (B) The need for a new central tendency in Phase 2 in BPD participants is further emphasised by examining the central tendency of prior beliefs about partners under model M4 for BPD participants (also see [Figure S4](#)). Displayed is the distribution of individual parameter estimates (β_{par}^m ; left and α_{par}^m ; right) for BPD participants in Phase 2. Prosocial-competitive beliefs (β) have a large shifted central tendency between self (red) and other prior beliefs (grey). In contrast, Individualistic beliefs (α) have a relatively smaller self-other discrepancy. (C) The statistical agreement in the estimation of β_{par}^{ref} (top) and α_{par}^{ref} (bottom) for both SEP and FULL M1 models. Distortion is minimal (but non-zero).



Supplementary Figure 11.

Pearson correlation between parameters of equivalence across models M1-M4.

All models were hierarchically fitted (using the HBM package; Piray et al., 2019) without comparison to each group. We then compared prior flexibility over others in M1 and M2 (which allow for self-insertion) to the absolute difference in median shift of beliefs between phase 1 and phase 2 under M3 and M4 (which do not allow for self-insertion). We also correlated prior flexibility over others in phase 2 estimated under M1 and M2, as well as central tendency over new priors over others under M3 and M4. We find excellent convergence between approximated parameters of equivalent meaning across models.

S1	O1	S2	O2	Self-Disparity	Other Disparity	Type
10	6	8	2	2	4	I-C
7	7	10	7	3	0	P-I
7	1	8	5	1	4	C-I
10	5	10	10	0	5	C-P
12	9	9	9	3	0	I-P
10	5	8	1	2	4	I-C
6	2	8	6	2	4	C-I
8	2	9	6	1	4	C-I
5	5	5	1	0	4	P-C
7	7	7	2	0	5	P-C
12	8	8	8	4	0	I-P
8	8	8	2	0	6	P-C
9	5	7	1	2	4	I-C
6	6	8	6	2	0	P-I
6	1	7	5	1	4	C-I
12	6	10	2	2	4	I-C
7	7	7	1	0	6	P-C
10	6	6	6	4	0	I-P
4	4	8	4	4	0	P-I
11	6	9	2	2	4	I-C
5	1	7	5	2	4	C-I
8	5	8	8	0	3	C-P
6	6	10	6	4	0	P-I
6	3	6	6	0	3	C-P
5	5	8	5	3	0	P-I
11	5	9	1	2	4	I-C
9	5	9	9	0	4	C-P
7	3	7	7	0	4	C-P
8	3	8	8	0	5	C-P
12	10	10	10	2	0	I-P
10	8	8	8	2	0	I-P
6	6	6	1	0	5	P-C
6	6	6	2	0	4	P-C
7	2	8	6	1	4	C-I
10	7	7	7	3	0	I-P
8	8	10	8	2	0	P-I

Supplementary Table 1.

Option pair rewards for each phase and their corresponding 'type'. Within phase order of trials were randomised. P=Prosocial, I=Individualistic, C=Competitive. S1 = reward to self for option 1. S2 = reward to self for option 2. O1 = reward to other for option 1. O2 = reward to other for option 2.

Model = M1

Parameter	BPD	CON	$\Delta\mu+90\%HDI$
<i>Directly Approximated Parameters</i>			
α_{ppt}^m	9.97[7.69]	16.2[8.62]	6.2[4.1, 8.3]
β_{ppt}^m	-6.81[5.77]	-7.26[5.46]	-0.4[-1.9, 1.0]
α_{ppt}^σ	0.75 [0.07]	2.49[3.16]	1.8 [1.2, 2.4]
β_{ppt}^σ	3.38[2.18]	4.65[7.79]	1.3 [-0.3, 2.8]
α_{par}^{ref}	6.23[4.23]	7.76[4.32]	1.5 [0.5, 2.6]
β_{par}^{ref}	10.38[6.15]	8.24[5.19]	-2.1[-3.5, -0.7]
<i>Derived Parameters</i>			
$\Delta\alpha_{par}^m$	5.21[4.06]	8.16[4.88]	2.9 [1.8, 4.1]
$\Delta\beta_{par}^m$	13.42[6.21]	16.41[8.35]	3.0 [1.2, 4.9]
α_{par}^σ	8.20[5.28]	8.24[5.24]	0 [-1.3, 1.3]
β_{par}^σ	7.09[5.79]	10.52[6.70]	3.4 [1.8, 5.0]
$ \Delta\alpha_{ppt}^m $	0.07[0.10]	0.69[1.61]	0.6[0.3, 0.9]
$ \Delta\beta_{ppt}^m $	1.42[2.00]	2.14[2.73]	0.7[0.1, 1.3]
$ \Delta\alpha_{ppt}^\sigma $	-0.07[0.09]	-0.77[1.89]	-0.7[-1.0, -0.4]
$ \Delta\beta_{ppt}^\sigma $	-1.34[1.94]	-1.58[2.08]	-0.2[-0.3, 0.7]

Model = M2

Parameter	BPD	CON	$\Delta\mu+90\%HDI$
<i>Directly Approximated Parameters</i>			
α_{ppt}^m	9.60[7.49]	15.7[8.28]	6.1[4.1, 8.1]
β_{ppt}^m	-5.89[5.22]	-6.13[4.91]	-0.2[-1.5, 1.1]
α_{ppt}^σ	0.68 [0.04]	2.74[1.64]	2.1 [1.8, 2.4]
β_{ppt}^σ	2.69[1.84]	3.03[4.41]	0.3 [-0.5, 1.2]
α_{par}^{ref}	6.00[4.13]	7.44[3.87]	1.4 [0.4, 2.4]
β_{par}^{ref}	10.90[6.63]	8.76[5.49]	-2.1[-3.7, -0.6]
<i>Derived Parameters</i>			
$\Delta\alpha_{par}^m$	4.53[4.01]	8.22[4.81]	3.7 [2.6, 4.8]
$\Delta\beta_{par}^m$	12.01[6.54]	15.30[8.23]	3.3 [1.4, 5.2]
α_{par}^σ	2.34[1.23]	2.81[1.32]	0.5 [0.1, 0.8]
β_{par}^σ	1.84[1.08]	2.76[1.82]	0.9 [0.6, 1.3]

Model = M3

Parameter	BPD	CON	$\Delta\mu+90\%HDI$
<i>Directly Approximated Parameters</i>			
α_{ppt}^m	10.14[7.95]	16.8[7.87]	6.7[4.7, 8.7]

Supplementary Table 2.

Model Parameters of M1-M4 Following Independent Hierarchical Fitting For All Participants.

β_{ppt}^m	-7.56[6.55]	-8.10[6.94]	-0.5[-2.2, 1.2]
α_{ppt}^σ	0.58 [0.06]	0.79[0.24]	0.2 [0.2, 0.3]
β_{ppt}^σ	2.48[1.26]	2.64[1.47]	0.2 [-0.2, 0.5]
α_{par}^m	7.19[4.87]	9.12[5.93]	1.1[0.5, 3.3]
β_{par}^m	-0.57[1.72]	0.43[3.40]	1.0[0.3, 1.7]
α_{par}^{ref}	1.24[0.85]	2.32[1.62]	1.1 [0.8, 1.4]
β_{par}^{ref}	6.29[4.14]	4.56[2.98]	-1.7[-2.6, -0.8]
<i>Derived Parameters</i>			
$\Delta\alpha_{par}^m$	0.29[0.24]	1.35[1.74]	1.1 [0.7, 1.4]
$\Delta\beta_{par}^m$	6.74[5.27]	8.90[6.45]	2.2 [0.7, 3.7]
α_{par}^σ	1.02[0.49]	1.63[0.84]	0.5 [0.4, 0.7]
β_{par}^σ	1.56[0.86]	2.15[1.73]	0.6 [0.2, 1.0]
$ \Delta\alpha_{ppt}^m $	1.53[1.42]	2.22[3.01]	0.7 [0.1, 1.3]
$ \Delta\beta_{ppt}^m $	3.03[3.88]	5.86[5.35]	2.8 [1.6, 4.0]
$ \Delta\alpha_{ppt}^\sigma $	-0.11[0.07]	-0.13[0.19]	-0.0[0.0, 0.1]
$ \Delta\beta_{ppt}^\sigma $	-0.58[0.97]	-0.77[1.75]	-0.2[0.2, -0.6]
Model = M4			
Parameter	BPD	CON	$\Delta\mu+90\%HDI$
<i>Directly Approximated Parameters</i>			
α_{ppt}^m	9.62[7.38]	15.51[7.66]	5.9[3.9, 7.8]
β_{ppt}^m	-5.99[5.34]	-6.11[5.26]	-0.1[-1.5, 1.2]
α_{ppt}^σ	0.74 [0.04]	2.08[1.17]	1.3 [1.1, 1.5]
β_{ppt}^σ	2.24[0.98]	2.23[1.38]	0.0 [-0.3, 0.3]
α_{par}^m	7.19[4.67]	9.54[5.63]	2.4[1.0, 3.7]
β_{par}^m	-1.29[1.27]	-1.49[0.59]	-0.2[-0.5, 0.0]
α_{par}^{ref}	1.86[1.68]	2.45[1.08]	0.6 [0.2, 0.9]
β_{par}^{ref}	7.55[4.57]	6.74[4.02]	-0.8[-1.9, -0.3]
<i>Derived Parameters</i>			
$\Delta\alpha_{par}^m$	1.46[2.04]	2.54[2.32]	1.5 [1.0, 2.1]
$\Delta\beta_{par}^m$	7.30[5.16]	10.90[6.48]	2.2 [0.7, 3.7]
α_{par}^σ	1.34[0.79]	1.84[0.67]	0.6 [0.4, 0.8]
β_{par}^σ	1.63[0.89]	2.36[1.66]	0.6 [0.2, 1.0]

Supplementary Table 2. (continued)

M1				
	Est.	se	t	p
Type $[\beta \text{ vs. } \alpha]$	0.21	0.019	10.99	<2e-16
Trial	-0.015	0.0001	-23.87	<2e-16
Type x Trial	-0.016	0.001	-13.36	<2e-16
Group $[D_{KL}(\alpha_{par}^m)]$	0.083	0.030	2.78	0.006
Group $[D_{KL}(\beta_{par}^m)]$	-0.064	0.028	-2.31	0.023
M2				
	Est.	se	t	p
Type $\beta \text{ vs. } \alpha$	0.21	0.019	10.92	<2e-16
Trial	-0.014	0.0001	-22.70	<2e-16
Type x Trial	-0.016	0.0001	-12.88	<2e-16
Group $[D_{KL}(\alpha_{par}^m)]$	0.090	0.030	3.02	0.003
Group $[D_{KL}(\beta_{par}^m)]$	-0.064	0.030	-2.16	0.033
M3				
	Est.	se	t	p
Type $\beta \text{ vs. } \alpha$	0.27	0.019	14.47	<2e-16
Trial	-0.0076	0.001	-11.52	<2e-16
Type x Trial	-0.014	0.001	-11.49	<2e-16
Group $[D_{KL}(\alpha_{par}^m)]$	0.254	0.051	4.92	3.37e-06
Group $[D_{KL}(\beta_{par}^m)]$	-0.360	0.050	-7.71	7.56e-12
M4				
	Est.	se	t	p
Type $\beta \text{ vs. } \alpha$	0.33	0.019	17.40	<2e-16
Trial	-0.018	0.0001	-17.61	<2e-16
Type x Trial	-0.019	0.001	-16.10	<2e-16
Group $[D_{KL}(\alpha_{par}^m)]$	0.120	0.040	2.94	0.004
Group $[D_{KL}(\beta_{par}^m)]$	-0.090	0.040	-2.17	0.032

Table T3

Random-effect linear relationships between DKL, trial, group, and preferences type for each model (M1-M4) following Independent Hierarchical Fitting For All Participants.

Estimates are the scaled change in D_{KL} as a result of each fixed effect. ID was used as a random variable to control for within-subject effects. Group effects (CON vs BPD) were analysed for the D_{KL} within each preference type.

Supplementary Text 1. Reaction times in the Intentions Game

Phase 1

Examining reaction times (in milliseconds; ms) in phase 1 by choice type revealed that, compared to competitive choices, individualistic choices were made faster (linear estimate = -880.60, 95%CI: -1385.42, -376.2; $t = -3.42$, $p < 0.001$), and prosocial choices were made fastest (linear estimate = -1171.1, 95%CI: -1701.97, -640.71; $t = -4.32$, $p < 0.001$) irrespective of the type of choice pair. Prosocial choices were made significantly faster than individualistic choices (linear estimate = -290.70, 95%CI: -548.50, -32.91; $t = -2.21$, $p = 0.027$).

Phase 2

All participants were slower at the start of phase 2 and sped up over time (linear estimate = -15.03, 95%CI: -21.06, -8.99; $t = -4.88$, $p < 0.001$). Baseline participant-partner similarity did not have an overall effect on reaction time but did interact with trial – as participant-partner similarity increased, reaction times early in phase 2 were significantly slower and this effect attenuated over time (linear estimate = -0.53, 95%CI: -0.75, -0.32; $t = -4.91$, $p < 0.001$; see **Figure S9**). Reaction time did not vary between groups: both BPD and CON participants displayed the same effect.

Reaction times and belief updates in phase 2 were significantly coupled, such that larger shifts in posterior beliefs along both axes were associated with larger reaction times (linear estimate [$D_{DL}(\alpha_{par}^m)$] = 0.044, 95%CI: 0.027, 0.06, $t = 5.01$, $p < 0.001$; linear estimate [$D_{KL}(\beta_{par}^m)$] = 0.021, 95%CI: 0.005, 0.039, $t = 2.49$, $p = 0.012$).

Phase 3

Reaction times in phase 3 revealed that compared to competitive choices, individualistic choices were made faster (linear estimate = -528.50, 95%CI: -943.60, -114.6; $t = -2.50$, $p = 0.012$), and prosocial choices were made fastest (linear estimate = -693.5, 95%CI: -1137.65, -250.39; $t = -3.07$, $p = 0.002$). Prosocial choices were no longer executed significantly faster than individualistic choices. All participants made faster choices in phase 3 compared to phase 1 (linear estimate = -242.02, 95% CI: -332.64, -151.41; $t = -5.24$, $p = 0.001$).

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Author information

Joseph M Barnby

Department of Psychology, Royal Holloway, University of London, London, United Kingdom,
School of Psychiatry and Clinical Neuroscience, University of Western Australia, Perth,
Australia

For correspondence: joseph.barnby@rhul.ac.uk

Jen Nguyen

Department of Psychology, Royal Holloway, University of London, London, United Kingdom

Julia Griem

Department for Clinical, Educational, and Health Psychology, Division of Psychology and
Language Sciences, University College London, London, United Kingdom, Anna Freud,
London, United Kingdom

Magdalena Wloszek

Anna Freud, London, United Kingdom

Henry Burgess

Department of Neuroscience, Washington University in St. Louis, St. Louis, United States

Linda Richards

Department of Neuroscience, Washington University in St. Louis, St. Louis, United States

Jessica Kingston

Department of Psychology, Royal Holloway, University of London, London, United Kingdom

Gavin Cooper

Department of Psychology, Royal Holloway, University of London, London, United Kingdom

London Personality and Mood Disorders Consortium

P Read Montague

Centre for Human Neuroscience Research, Virginia Tech, Roanoke, United States

Peter Dayan

Max Planck Institute of Biological Cybernetics, Tübingen, Germany, University of Tübingen,
Tübingen, Germany

Tobias Nolte

Department for Clinical, Educational, and Health Psychology, Division of Psychology and
Language Sciences, University College London, London, United Kingdom, Anna Freud,
London, United Kingdom

Peter Fonagy

Department for Clinical, Educational, and Health Psychology, Division of Psychology and
Language Sciences, University College London, London, United Kingdom, Anna Freud,

London, United Kingdom

Editors

Reviewing Editor

Roshan Cools

Donders Institute for Brain, Cognition and Behaviour, Radboud University Nijmegen, Nijmegen, Netherlands

Senior Editor

Michael Frank

Brown University, Providence, United States of America

Reviewer #1 (Public review):

Summary:

The authors use a sophisticated task design and Bayesian computational modeling to test their hypothesis that information generalization (operationalized as a combination of self-insertion and social contagion) in social situations is disrupted in Borderline Personality Disorder. Their main finding relates to the observation that two different models best fit the two tested groups: While the model assuming both self-insertion and social contagion to be present when estimating others' social value preferences fit the control group best, a model assuming neither of these processes provided the best fit to BPD participants.

Strengths:

The revisions have substantially strengthened the paper and the manuscript is much clearer and easier to follow now. The strengths of the presented work lie in the sophisticated task design and the thorough investigation of their theory by use of mechanistic computational models to elucidate social decision-making and learning processes in BPD.

Weaknesses:

Some critical concerns remain after the first revision, particularly regarding the use of causal language and the clarity of the hypotheses and results, specified in the points below.

(1) The authors frequently refer to their predictions and theory as being causal, both in the manuscript and in their response to reviewers. However, causal inference requires careful experimental design, not just statistical prediction. For example, the claim that "algorithmic differences between those with BPD and matched healthy controls" are "causal" in my opinion is not warranted by the data, as the study does not employ experimental manipulations or interventions which might predictably affect parameter values. Even if model parameters can be seen as valid proxies to latent mechanisms, this does not automatically mean that such mechanisms cause the clinical distinction between BPD and CON, they could plausibly also refer to the effects of therapy or medication. I recommend that such causal language, also implicit to expressions like "parameter influences on explicit intentional attributions", is toned down throughout the manuscript.

(2) Although the authors have now much clearer outlined the study's aims, there still is a lack of clarity with respect to the authors' specific hypotheses. I understand that their primary predictions about disruptions to self-other generalisation processes underlying BPD are embedded in the four main models that are tested, but it is still unclear what specific hypotheses the authors had about group differences with respect to the tested models. I recommend the authors specify this in the introduction rather than referring to prior work where the same hypotheses may have been mentioned.

(3) Caveats should also be added about the exploratory nature of the many parameter group comparisons. If there are any predictions about group differences that can be made based on prior literature, the authors should make such links clear.

(4) I'm not sure I understand why the authors, after adding multiple comparison correction, now list two kinds of p-values. To me, this is misleading and precludes the point of multiple comparison corrections, I therefore recommend they report the FDR-adjusted p-values only. Likewise, if a corrected p-value is greater than 0.05 this should not be interpreted as a result.

(5) Can the authors please elaborate why the algorithm proposed to be employed by BPD is more 'entropic', especially given both their self-priors and posteriors about partners' preferences tended to be more precise than the ones used by CON? As far as I understand, there's nothing in the data to suggest BPD predictions should be more uncertain. In fact, this leads me to wonder, similarly to what another reviewer has already suggested, whether BPD participants generate self-referential priors over others in the same way CON participants do, they are just less favourable (i.e., in relation to oneself, but always less prosocial) - I think there is currently no model that would incorporate this possibility? It should at least be possible to explore this by checking if there is any statistical relationship between the estimated θ_{ppt}^m and $p(\theta_{par} | D^0)$.

"To note, social contagion under M3 was highly correlated with contagion under M1 (see Fig S11). This provides some preliminary evidence that trauma impacts beliefs about individualism directly, whereas trauma and persecutory beliefs impact beliefs about prosociality through impaired trait mentalising" - I don't understand what the authors mean by this, can they please elaborate and add some explanation to the main text?

<https://doi.org/10.7554/eLife.104008.2.sa3>

Reviewer #2 (Public review):

Summary:

The paper investigates social-decision making, and how this changes after observing the behaviour of other people, in borderline personality disorder. The paper employs a task including three phases, the first where participants make decision on how to allocate rewards to oneself and to a virtual partner, the second where they observe the same task performed by someone else, and a third phase equivalent to phase one, but with a new partner. Using sophisticated computational modelling to analyse choice data, the study reports that borderline participants (versus controls) are more certain about their preferences in phase one, used more neutral priors and are less flexible during phase two, and are less influenced by partners in phase three.

Strengths:

The topic is interesting and important, and the findings are potentially intriguing. The computational methods employed is clever and sophisticated, at the cutting edge of research in the field.

Weaknesses:

The paper is not based on specific empirical hypotheses formulated at the outset, but, rather, it uses an exploratory approach. Indeed, the task is not chosen in order to tackle specific empirical hypotheses. This, in my view, is a limitation since the introduction reads a bit vague and it is not always clear which gaps in the literature the paper aims to fill. As a further consequence, it is not always clear how the findings speak to previous theories on the topic.

Reviewer #3 (Public review):

In this paper, the authors use a three-phase economic game to examine the tendency to engage in prosocial versus competitive exchanges with three anonymous partners. In particular, they consider individual differences in the tendency to infer about others' tendencies based on one's preferences and to update one's preferences based on observations of others' behavior. The study includes a sample of individuals diagnosed with borderline personality disorder and a matched sample of psychiatrically healthy control participants.

On the whole, the experimental design is well-suited to the questions and the computational model analyses are thorough, including modern model-fitting procedures. I particularly appreciated the clear exposition regarding model parameterization and the descriptive Table 2 for qualitative model comparison. In the revised manuscript, the authors now provide a more thorough treatment of examining group differences in computational parameters given that the best-fitting model differed by group. They also examine the connection of their task and findings to related research focusing on self-other representation and mentalization (e.g., Story et al., 2024).

The authors note that the task does not encourage competition and instead captures individual differences in the motivation to allocate rewards to oneself and others in an interdependent setting. The paper could have been strengthened by clarifying how the Social Value Orientation framework can be used to interpret the motivations and behavior of BPD versus CON participants on the task. Although the authors note that their approach makes "clear and transparent a priori predictions," the paper could be improved by providing a clear and consolidated statement of these predictions so that the results could be interpreted vis-a-vis any a priori hypotheses.

Finally, the authors have amended their individual difference analyses to examine psychometric measures such as the CTQ alongside computational model parameter estimate differences. I appreciate that these analyses are described as exploratory. The approach of using a partial correlation network with bootstrapping (and permutation) was interesting, but the logic of the analysis was not clearly stated. In particular, there are large group (Table 1: CON vs. BPD) differences in the measures introduced into this network. As a result, it is hard to understand whether any partial correlations are driven primarily by mean differences in severity (correlations tend to be inflated in extreme groups designs due to the absence of observation in middle of scales forming each bivariate distribution). I would have found these exploratory analyses more revealing if group membership was controlled for.

<https://doi.org/10.7554/eLife.104008.2.sa1>

Author response:

The following is the authors' response to the original reviews

Response to the Editors' Comments

Thankyou for this summary of the reviews and recommendations for corrections. We respond to each in turn, and have documented each correction with specific examples contained within our response to reviewers below.

They all recommend to clarify the link between hypotheses and analyses, ground them more clearly in, and conduct critical comparisons with existing literature, and address a

potential multiple comparison problem.'

We have restructured our introduction to include the relevant literature outlined by the reviewers, and to be more clearly ground the goals of our model and broader analysis. We have additionally corrected for multiple comparisons within our exploratory associative analyses. We have additionally sign posted exploratory tests more clearly.

'Furthermore, R1 also recommends to include a formal external validation of how the model parameters relate to participant behaviour, to correct an unjustified claim of causality between childhood adversity and separation of self, and to clarify role of therapy received by patients.'

We have now tempered our language in the abstract which unintentionally implied causality in the associative analysis between childhood trauma and other-to-self generalisation. To note, in the sense that our models provide causal explanations for behaviour across all three phases of the task, we argue that our model comparison provides some causal evidence for algorithmic biases within the BPD phenotype. We have included further details of the exclusion and inclusion criteria of the BPD participants within the methods.

R2 specifically recommends to clarify, in the introduction, the specific aim of the paper, what is known already, and the approach to addressing it.'

We have more thoroughly outlined the current state of the art concerning behavioural and computational approaches to self insertion and social contagion, in health and within BPD. We have linked these more clearly to the aims of the work.

'R2 also makes various additional recommendations regarding clarification of missing information about model comparison, fit statistics and group comparison of parameters from different models.'

Our model comparison approach and algorithm are outlined within the original paper for Hierarchical Bayesian Model comparison (Piray et al., 2019). We have outlined the concepts of this approach in the methods. We have now additionally improved clarity by placing descriptions of this approach more obviously in the results, and added points of greater detail in the methods, such as which statistics for comparison we extracted on the group and individual level.

In addition, in response to the need for greater comparison of parameters from different models, we have also hierarchically force-fitted the full suite of models (M1-M4) to all participants. We report all group differences from each model individually – assuming their explanation of the data - in Table S2. We have also demonstrated strong associations between parameters of equivalent meaning from different models to support our claims in Fig S11. Finally, we show minimal distortion to parameter estimates in between-group analysis when models are either fitted hierarchically to the entire population, or group wise (Figure S10).

'R3 additionally recommends to clarify the clinical and cognitive process relevance of the experiment, and to consider the importance of the Phase 2 findings.'

We have now included greater reference to the assumptions in the social value orientation paradigm we use in the introduction. We have also responded to the specific point about the shift in central tendencies in phase 2 from the BPD group, noting that, while BPD participants do indeed get more *relatively* competitive vs. CON participants, they remain strikingly neutral with respect to the overall statespace. Importantly, model M4 does not preclude more competitive distributions existing.

'Critically, they also share a concern about analyzing parameter estimates fit separately to two groups, when the best-fitting model is not shared. They propose to resolve this by considering a model that can encompass the full dynamics of the entire sample.'

We have hierarchically force-fitted the full suite of models (M1-M4) to all participants to allow for comparison between parameters within each model assumption. We report all group differences from each model individually – assuming their explanation of the data – in Table S2 and Table S3. We have also demonstrated strong associations between parameters of equivalent meaning from different models to support our claims in Fig S11. We also show minimal distortion to parameter estimates in between-group analysis when models are either fitted hierarchically to the entire population, or group wise (Figure S10).

Within model M1 and M2, the $\alpha_{par}^{ref}, \beta_{par}^{ref}$ parameters quantify the degree to which participants believe their partner to be different from themselves. Under M1 and M2 model assumptions, BPD participants have meaningfully larger β_{par}^{ref} versus CON (Fig S10), which supports the notion that a new central tendency may be more parsimonious in phase 2 (as in the case of the optimal model for BPD, M4). We also show strong correlations across models between $\alpha_{par}^{ref}, \beta_{par}^{ref}$ under M1 and M2, and the shift in central tendencies of beliefs between phase 1 and 2 under M3 and M4. This supports our primary comparison, and shows that even under non-dominant model assumptions, parameters demonstrate that BPD participants expect their partner's relative reward preferences to be vastly different from themselves versus CON.

'A final important point concerns the psychometric individual difference analyses which seem to be conducted on the full sample without considering the group structure.'

We have now more clearly focused our psychometric analysis. We control for multiple comparisons, and compare parameters across the same model (M3) when assessing the relationship between paranoia, trauma, trait mentalising, and social contagion. We have relegated all other exploratory analyses to the supplementary material and noted where p values survive correction using False Discovery Rate.

Reviewer 1:

'The manuscript's primary weakness relates to the number of comparisons conducted and a lack of clarity in how those comparisons relate to the authors' hypotheses. The authors specify a primary prediction about disruption to information generalization in social decision making & learning processes, and it is clear from the text how their 4 main models are supposed to test this hypothesis. With regards to any further analyses however (such as the correlations between multiple clinical scales and eight different model parameters, but also individual parameter comparisons between groups), this is less clear. I recommend the authors clearly link each test to a hypothesis by specifying, for each analysis, what their specific expectations for conducted comparisons are, so a reader can assess whether the results are/aren't in line with predictions. The number of conducted tests relating to a specific hypothesis also determines whether multiple comparison corrections are warranted or not. If comparisons are exploratory in nature, this should be explicitly stated.'

We have now corrected for multiple comparisons when examining the relationship between psychometric findings and parameters, using partial correlations and bootstrapping for

robustness. These latter analyses were indeed not preregistered, and so we have more clearly signposted that these tests were exploratory. We chose to focus on the influence of psychometrics of interest on social contagion under model M3 given that this model explained a reasonable minority of behaviour in each group. We have now fully edited this section in the main text in response, and relegated all other correlations to the supplementary materials.

'Furthermore, the authors present some measures for external validation of the models, including comparison between reaction times and belief shifts, and correlations between model predicted accuracy and behavioural accuracy/total scores. However it would be great to see some more formal external validation of how the model parameters relate to participant behaviour, e.g., the correlation between the number of pro-social choices and β -values, or the correlation between the change in absolute number of pro-social choices and the change in β . From comparing the behavioural and computational results it looks like they would correlate highly, but it would be nice to see this formally confirmed.'

We have included this further examination within the Generative Accuracy and Recovery section:

'We also assessed the relationship (Pearson r s) between modelled participant preference parameters in phase 1 and actual choice behaviour: β_{ppt}^m was negatively correlated with prosocial versus competitive choices ($r=-0.77$, $p<0.001$) and individualistic versus competitive choices ($r=-0.59$, $p<0.001$); α_{ppt}^m was positively correlated with individualistic versus competitive choices ($r=0.53$, $p<0.001$) and negatively correlated with prosocial versus individualistic choices ($r=-0.69$, $p<0.001$).'

'The statement in the abstract that 'Overall, the findings provide a clear explanation of how self-other generalisation constrains and assists learning, how childhood adversity disrupts this through separation of internalised beliefs' makes an unjustified claim of causality between childhood adversity and separation of self - and other beliefs, although the authors only present correlations. I recommend this should be rephrased to reflect the correlational nature of the results.'

Sorry – this was unfortunate wording: we did not intend to imply causation with our second clause in the sentence mentioned. We have amended the language to make it clear this relationship is associative:

'Overall, the findings provide a clear explanation of how self-other generalisation constrains and assists learning, how childhood adversity is associated with separation of internalised beliefs, and makes clear causal predictions about the mechanisms of social information generalisation under uncertainty.'

'Currently, from the discussion the findings seem relevant in explaining certain aberrant social learning and -decision making processes in BPD. However, I would like to see a more thorough discussion about the practical relevance of their findings in light of their observation of comparable prediction accuracy between the two groups.'

We have included a new paragraph in the discussion to address this:

'Notably, despite differing strategies, those with BPD achieved similar accuracy to CON participants in predicting their partners. All participants were more concerned with relative

versus absolute reward; only those with BPD changed their strategy based on this focus. Practically this difference in BPD is captured either through disintegrated priors with a new median (M4) or very noisy, but integrated priors over partners (M1) if we assume M1 can account for the full population. In either case, the algorithm underlying the computational goal for BPD participants is far higher in entropy and emphasises a less stable or reliable process of inference. In future work, it would be important to assess this mechanism alongside momentary assessments of mood to understand whether more entropic learning processes contribute to distressing mood fluctuation.'

'Relatedly, the authors mention that a primary focus of mentalization based therapy for BPD is 'restoring a stable sense of self' and 'differentiating the self from the other'. These goals are very reminiscent of the findings of the current study that individuals with BPD show lower uncertainty over their own and relative reward preferences, and that they are less susceptible to social contagion. Could the observed group differences therefore be a result of therapy rather than adverse early life experiences?'

This is something that we wish to explore in further work. While verbal and model descriptions appear parsimonious, this is not straight forward. As we see, clinical observation and phenomenological dynamics may not necessarily match in an intuitive way to parameters of interest. It may be that compartmentalisation of self and other – as we see in BPD participants within our data – may counter-intuitively express as a less stable self. The evolutionary mechanisms that make social insertion and contagion enduring may also be the same that foster trust and learning.

'Regarding partner similarity: It was unclear to me why the authors chose partners that were 50% similar when it would be at least equally interesting to investigate self-insertion and social contagion with those that are more than 50% different to ourselves? Do the authors have any assumptions or even data that shows the results still hold for situations with lower than 50% similarity?'

While our task algorithm had a high probability to match individuals who were approximately 50% different with respect to their observed behaviour, there was variation either side of this value. The value of 50% median difference was chosen for two reasons: 1. We wanted to ensure participants had to learn about their partner to some degree relative to their own preferences and 2. we did not want to induce extreme over or under familiarity given the (now replicated) relationship between participant-partner similarity and intentional attributions (see below). Nevertheless, we did have some variation around the 50% median. Figure 3A in the top left panel demonstrates this fluctuation in participant-partner similarity and the figure legend further described this distribution (mean = 49%, sd = 12%). In future work we want to more closely manipulate the median similarity between participants and partners to understand how this facilitates or inhibits learning and generalisation.

There is some analysis of the relationship between degrees of similarity and behaviour. In the third paragraph of page 15 we report the influence of participant-partner similarity on reaction times. In prior work (Barnby et al., 2022; Cognition) we had shown that similarity was associated with reduced attributions of harm about a partner, irrespective of their true parameters (e.g. whether they were prosocial/competitive). We replicate this previous finding with a double dissociation illustrated in Figure 4, showing that greater discrepancies in participant-partner prosociality increases explicit harmful intent attributions (but not self-interest), and discrepancies in participant-partner individualism reduces explicit self-interest attributions (but not harmful intent). We have made these clearer in our results structure, and included FDR correction values for multiple comparisons.

The methods section is rather dense and at least I found it difficult to keep track of the many different findings. I recommend the authors reduce the density by moving some of the secondary analyses in the supplementary materials, or alternatively, to provide an overall summary of all presented findings at the end of the Results section.

We have now moved several of our exploratory findings into the supplementary materials, notably the analysis of participant-partner similarity on reaction times (Fig S9), as well as the uncorrected correlation between parameters (Fig S7).

Fig 2C) and Discussion p. 21: What do the authors mean by 'more sensitive updates'? more sensitive to what?

We have now edited the wording to specify 'more belief updating' rather than 'sensitive' to be clearer in our language.

P14 bottom: please specify what is meant by axial differences.

We have changed this to 'preference type' rather than using the term 'axial'.

It may be helpful to have Supplementary Figure 1 in the main text.

Thank you for this suggestion. Given the volume of information in the main text we hope that it is acceptable for Figure S1 to remain in the supplementary materials.

Figure 3D bottom panel: what is the difference between left and right plots? Should one of them be alpha not beta?

The left and right plots are of the change in standard deviation (left) and central tendency (right) of participant preference change between phase 1 and 3. This is currently noted in the figure legend, but we had added some text to be clearer that this is over prosocial-competitive beliefs specifically. We chose to use this belief as an example given the centrality of prosocial-competitive beliefs in the learning process in Figure 2. We also noticed a small labelling error in the bottom panels of 3D which should have noted that each plot was either with respect to the precision or mean-shift in beliefs during phase 3.

'The relationship between uncertainty over the self and uncertainty over the other with respect to the change in the precision (left) and median-shift (right) in phase 3 prosocial-competitive beliefs (β_{ppt}).'

Supplementary Figure 4: The prior presented does not look neutral to me, but rather right-leaning, so competitive, and therefore does indeed look like it was influenced by the self-model? If I am mistaken please could the authors explain why.

This example distribution is taken from a single BPD participant. In this case, indeed, the prior is somewhat right-shifted. However, on a group level, priors over the partner were closely centred around 0 (see reported statistics in paragraph 2 under the heading 'Phase 2 – BPD Participants Use Disintegrated and Neutral Priors'). However, we understand how this may come across as misleading. For clarity we have expanded upon Figure S4 to include the phase 1 and prior phase 2 distributions for the entire BPD population for both prosocial and individualistic beliefs. This further demonstrates that those with BPD held surprisingly neutral beliefs over the expectations about their partners' prosociality, but had minor shifts

between their own individualistic preferences and the expected individualistic preferences of their partners. This is also visible in Figure S2.

Reviewer 2:

'There are two major weaknesses. First, the paper lacks focus and clarity. The introduction is rather vague and, after reading it, I remained confused about the paper's aims. Rather than relying on specific predictions, the analysis is exploratory. This implies that it is hard to keep track, and to understand the significance, of the many findings that are reported.'

Thank you for this opportunity to be clearer in our framing of the paper. While the model makes specific causal predictions with respect to behavioural dynamics conditional on algorithmic differences, our other analyses were indeed exploratory. We did not preregister this work but now given the intriguing findings we intent to preregister our future analyses.

We have made our introduction clearer with respect to the aims of the paper:

'Our present work sought to achieve two primary goals: 1. Extend prior causal computational theories to formalise the interrelation between self-insertion and social contagion within an economic paradigm, the Intentions Game and 2., Test how a diagnosis of BPD may relate to deficits in these forms of generalisation. We propose a computational theory with testable predictions to begin addressing this question. To foreshadow our results, we found that healthy participants employ a mixed process of self-insertion and contagion to predict and align with the beliefs of their partners. In contrast, individuals with BPD exhibit distinct, disintegrated representations of self and other, despite showing similar average accuracy in their learning about partners. Our model and data suggest that the previously observed computational characteristics in BPD, such as reduced self-anchoring during ambiguous learning and a relative impermeability of the self, arise from the failure of information about others to transfer to and inform the self. By integrating separate computational findings, we provide a foundational model and a concise, dynamic paradigm to investigate uncertainty, generalization, and regulation in social interactions.'

'Second, although the computational approach employed is clever and sophisticated, there is important information missing about model comparison which ultimately makes some of the results hard to assess from the perspective of the reader.'

Our model comparison employed what is state of the art random-effects Bayesian model comparison (Piray et al., 2019; PLOS Comp. Biol.). It initially fits each individual to each model using Laplace approximation, and subsequently 'races' each model against each other on the group level and individual level through hierarchical constraints and random-effect considerations. We included this in the methods but have now expanded on the description we used to compare models:

In the results -

'All computational models were fitted using a Hierarchical Bayesian Inference (HBI) algorithm which allows hierarchical parameter estimation while assuming random effects for group and individual model responsibility (Piray et al., 2019; see Methods for more information). We report individual and group-level model responsibility, in addition to protected exceedance probabilities between-groups to assess model dominance.'

We added to our existing description in the methods -

'All computational models were fitted using a Hierarchical Bayesian Inference (HBI) algorithm which allows hierarchical parameter estimation while assuming random effects

for group and individual model responsibility (Piray et al., 2019). During fitting we added a small noise floor to distributions ($2.22e^{-16}$) before normalisation for numerical stability. Parameters were estimated using the HBI in untransformed space drawing from broad priors ($\mu\mathbf{M}=0$, $\sigma^2_{\mathbf{M}} = 6.5$; where $\mathbf{M}=\{M1, M2, M3, M4\}$). This process was run independently for each group. Parameters were transformed into model-relevant space for analysis. All models and hierarchical fitting was implemented in Matlab (Version R2022B). All other analyses were conducted in R (version 4.3.3; arm64 build) running on Mac OS (Ventura 13.0). We extracted individual and group level responsibilities, as well as the protected exceedance probability to assess model dominance per group.'

(1) P3, third paragraph: please define self-insertion

We have now more clearly defined this in the prior paragraph when introducing concepts.

'To reduce uncertainty about others, theories of the relational self (Anderson & Chen, 2002) suggest that people have available to them an extensive and well-grounded representation of themselves, leading to a readily accessible initial belief (Allport, 1924; Kreuger & Clement, 1994) that can be projected or integrated when learning about others (self-insertion).'

(2) Introduction: the specific aim of the paper should be clarified - at the moment, it is rather vague. The authors write: "However, critical questions remain: How do humans adjudicate between self-insertion and contagion during interaction to manage interpersonal generalization? Does the uncertainty in self-other beliefs affect their generalizability? How can disruptions in interpersonal exchange during sensitive developmental periods (e.g., childhood maltreatment) inform models of psychiatric disorders?". Which of these questions is the focus of the paper? And how does the paper aim at addressing it?

(3) Relatedly, from the introduction it is not clear whether the goal is to develop a theory of self-insertion and social contagion and test it empirically, or whether it is to study these processes in BPD, or both (or something else). Clarifying which specific question(s) is addressed is important (also clarifying what we already know about that specific question, and how the paper aims at elucidating that specific question).

We have now included our specific aims of the paper. We note this in the above response to the reviewers general comments.

(4) "Computational models have probed social processes in BPD, linking the BPD phenotype to a potential over-reliance on social versus internal cues (Henco et al., 2020), 'splitting' of social latent states that encode beliefs about others (Story et al., 2023), negative appraisal of interpersonal experiences with heightened self-blame (Mancinelli et al., 2024), inaccurate inferences about others' irritability (Hula et al., 2018), and reduced belief adaptation in social learning contexts (Siegel et al., 2020). Previous studies have typically overlooked how self and other are represented in tandem, prompting further investigation into why any of these BPD phenotypes manifest." Not clear what the link between the first and second sentence is. Does it mean that previous computational models have focused exclusively on how other people are represented in BPD, and not on how the self is represented? Please spell this out.

Thank you for the opportunity to be clearer in our language. We have now spelled out our point more precisely, and included some extra relevant literature helpfully pointed out by another reviewer.

'Computational models have probed social processes in BPD, although almost exclusively during observational learning. The BPD phenotype has been associated with a potential over-

reliance on social versus internal cues (Henco et al., 2020), ‘splitting’ of social latent states that encode beliefs about others (Story et al., 2023), negative appraisal of interpersonal experiences with heightened self-blame (Mancinelli et al., 2024), inaccurate inferences about others’ irritability (Hula et al., 2018), and reduced belief adaptation in social learning contexts (Siegel et al., 2020). Associative models have also been adapted to characterize ‘leaky’ self-other reinforcement learning (Ereira et al., 2018), finding that those with BPD overgeneralize (leak updates) about themselves to others (Story et al., 2024). Altogether, there is currently a gap in the direct causal link between insertion, contagion, and learning (in)stability.’

(5) P5, first paragraph. The description of the task used in phase 1 should be more detailed. The essential information for understanding the task is missing.

We have updated this section to point toward Figure 1 and the Methods where the details of the task are more clearly outlined. We hope that it is acceptable not to explain the full task at this point for brevity and to not interrupt the flow of the results.

“Detailed descriptions of the task can be found in the methods section and Figure 1.”

(6) P5, second paragraph: briefly state how the Psychometric data were acquired (e.g., self-report).

We have now clarified this in the text.

‘All participants also self-reported their trait paranoia, childhood trauma, trust beliefs, and trait mentalizing (see methods).’

(7) "For example, a participant could make prosocial (self=5; other=5) versus individualistic (self=10; other=5) choices, or prosocial (self=10; other=10) versus competitive (self=10; other=5) choices". Not sure what criteria are used for distinguishing between individualistic and competitive - they look the same?

Sorry. This paragraph was not clear that the issue is that the interpretation of the choice depends on both members of the pair of options. Here, in one pair {(self=5,other=5) vs (self=10,other=5)}, it is highly pro-social for the self to choose (5,5), sacrificing 5 points for the sake of equality. In the second pair {(self=10,other=10) vs (self=10,other=5)}, it is highly competitive to choose (10,5), denying the other 5 points at no benefit to the self. We have clarified this:

‘We analyzed the ‘types’ of choices participants made in each phase (Supplementary Table 1). The interpretation of a participant’s choice depends on both values in a choice. For example, a participant could make prosocial (self=5; other=5) versus individualistic (self=10; other=5) choices, or prosocial (self=10; other=10) versus competitive (self=10; other=5) choices. There were 12 of each pair in phases 1 and 3 (individualistic vs. prosocial; prosocial vs. competitive; individualistic vs. competitive).’

(8) "In phase 1, both CON and BPD participants made prosocial choices over competitive choices with similar frequency (CON=9.67[3.62]; BPD=9.60[3.57])" please report t-test - the same applies also various times below.

We have now included the t test statistics with each instance.

‘In phase 3, both CON and BPD participants continued to make equally frequent prosocial versus competitive choices (CON=9.15[3.91]; BPD=9.38[3.31]; $t=-0.54$, $p=0.59$); CON participants continued to make significantly less prosocial versus individualistic choices (CON=2.03[3.45];

BPD=3.78 [4.16]; $t=2.31$, $p=0.02$). Both groups chose equally frequent individualistic versus competitive choices (CON=10.91[2.40]; BPD=10.18[2.72]; $t=-0.49$, $p=0.62$).’

(9) P 9: *"Models M2 and M3 allow for either self-insertion or social contagion to occur independently" what's the difference between M2 and M3?*

Model M2 hypothesises that participants use their own self representation as priors when learning about the other in phase 2, but are not influenced by their partner. M3 hypothesises that participants form an uncoupled prior (no self-insertion) about their partner in phase 2, and their choices in phase 3 are influenced by observing their partner in phase 2 (social contagion). In Figure 1 we illustrate the difference between M2 and M3. In Table 1 we specifically report the parameterisation differences between M2 and M3. We have also now included a correlational analysis of parameters between models to demonstrate the relationship between model parameters of equivalent value between models (Fig S11). We have also force fitted all models (M1-M4) to the data independently and reported group differences within each (see Table S2 and Table S3).

(10) P 9, last paragraph: *I did not understand the description of the Beta model.*

The beta model is outlined in detail in Table 1. We have also clarified the description of the beta model on page 9:

‘The ‘Beta model’ is equivalent to M1 in its causal architecture (both self-insertion and social contagion are hypothesized to occur) but differs in richness: it accommodates the possibility that participants might only consider a single dimension of relative reward allocation, which is typically emphasized in previous studies (e.g., Hula et al., 2018).’

(11) P 9: *I wonder whether one could think about more intuitive labels for the models, rather than M1, M2 etc.. This is just a suggestion, as I am not sure a short label would be feasible here.*

Thank you for this suggestion. We apologise that it is not very intuitive. The problem is that given the various terms we use to explain the different processes of generalisation that might occur between self and other, and given that each model is a different combination of each, we felt that numbering them was a lesser evil. We hope that the reader will be able to reference both Figure 1 and Table 1 to get a good feel for how the models and their causal implications differ.

(12) *Model comparison: the information about what was done for model comparison is scant, and little about fit statistics is reported. At the moment, it is hard for a reader to assess the results of the model comparison analysis.*

Model comparison and fitting was conducted using simultaneous hierarchical fitting and random-effects comparison. This is employed through the HBI package (Piray et al., 2019) where the assumptions and fitting procedures are outlined in great detail. In short, our comparison allows for individual and group-level hierarchical fitting and comparison. This overcomes the issue of interdependence between and within model fitting within a population, which is often estimated separately.

We have outlined this in the methods, although appreciate we do not touch upon it until the reader reaches that point. We have added a clarification statement on page 9 to rectify this:

‘All computational models were fitted using a Hierarchical Bayesian Inference (HBI) algorithm which allows hierarchical parameter estimation while assuming random effects for group and individual model responsibility (Piray et al., 2019; see Methods for more

information). We report individual and group-level model responsibility, in addition to protected exceedance probabilities between-groups to assess model dominance.'

(13) P 14, first paragraph: "BPD participants were also more certain about both types of preference" what are the two types of preferences?

The two types of preferences are relative (prosocial-competitive) and absolute (individualistic) reward utility. These are expressed as b and a respectively. We have expanded the sentence in question to make this clearer:

'BPD participants were also more certain about both self-preferences for absolute and relative reward ($\Delta\mu[\alpha_{ppt}^{\sigma}] = -0.89$, 95%HDI: -1.01, -0.75; $\Delta\mu[\beta_{ppt}^{\sigma}] = -0.32$, 95%HDI: -0.60, -0.04) versus CON participants (Figure 2B).'

(14) "Parameter Associations with Reported Trauma, Paranoia, and Attributed Intent" the results reported here are intriguing, but not fully convincing as there is the problem of multiple comparisons. The combinations between parameters and scales are rather numerous. I suggest to correct for multiple comparisons and to flag only the findings that survive correction.

We have now corrected this and controlled for multiple comparisons through partial correlation analysis, bootstrapping assessment for robustness, permutation testing, and False Detection Rate correction. We only report those that survive bootstrapping and permutation testing, reporting both corrected ($p[\text{fdr}]$) and uncorrected (p) significance.

(15) Results page 14 and page 15. The authors compare the various parameters between groups. I would assume that these parameters come from M1 for controls and from M4 for BDP? Please clarify if this is indeed the case. If it is the case, I am not sure this is appropriate. To my knowledge, it is appropriate to compare parameters between groups only if the same model is fit to both groups. If two different models are fit to each group, then the parameters are not comparable, as the parameter have, so to speak, different "meaning" in two models. Now, I want to stress that my knowledge on this matter may be limited, and that the authors' approach may be sound. However, to be reassured that the approach is indeed sound, I would appreciate a clarification on this point and a reference to relevant sources about this approach.

This is an important point. First, we confirmed all our main conclusions about parameter differences using the maximal model M1 to fit all the participants. We added Supplementary Table 2 to report the outcome of this analysis. Second, we did the same for parameters across all models M1-M4, fitting each to participants without comparison. This is particularly relevant for M3, since at least a minority of participants of both groups were best explained by this model. We report these analyses in Fig S11:

Since the M4 is nested within M1, we argue that this comparison is still meaningful, and note explanations in the text for why the effects noted between groups may occur given the differences in their causal meaning, for example in the results under phase 2 analyses:

'Belief updating in phase 2 was less flexible in BPD participants. Median change in beliefs (from priors to posteriors) about a partner's preferences was lower versus. CON ($= -5.53$, 95%HDI: -7.20, -4.00; $= -10.02$, 95%HDI: -12.81, -7.30). Posterior beliefs about partner were more precise in BPD versus CON ($= -0.94$, 95%HDI: -1.50, -0.45; $= -0.70$, 95%HDI: -1.20, -0.25). This is unsurprising given the disintegrated priors of the BPD group in M4, meaning they need to 'travel less' in state space. Nevertheless, even under assumptions of M1 and M2 for

both groups, BPD showed smaller posteriors median changes versus CON in phase 2 (see Table T2). These results converge to suggest those with BPD form rigid posterior beliefs.'

(16) *"We built and tested a theory of interpersonal generalization in a population of matched participants" this sentence seems to be unwarranted, as there is no theory in the paper (actually, as it is now, the paper looks rather exploratory)*

We thank the reviewer for their perspective. Formal models can be used as a theoretical statement on the casual algorithmic process underlying decision making and choice behaviour; the development of formal models are an essential theoretical tool for precision and falsification (Haslbeck et al., 2022). In this sense, we have built several competing formal theories that test, using casual architectures, whether the latent distribution(s) that generate one's choices generalise into one's predictions about another person, and simultaneously whether one's latent distribution(s) that represent beliefs about another person are used to inform future choices.

Reviewer 3:

'My broad question about the experiment (in terms of its clinical and cognitive process relevance): Does the task encourage competition or give participants a reason to take advantage of others? I don't think it does, so it would be useful to clarify the normative account for prosociality in the introduction (e.g., some of Robin Dunbar's work).'

We agree that our paradigm does not encourage competition. We use a reward structure that makes it contingent on participants to overcome a particular threshold before earning rewards, but there is no competitive element to this, in that points earned or not earned by partners have no bearing on the outcomes for the participant. This is important given the consideration of recursive properties that arise through mixed-motive games; we wanted to focus purely on observational learning in phase 2, and repercussion-free choices made by participants in phase 1 and 3, meaning the choices participants, and decisions of a partner, are theoretically in line with self-preferences irrespective of the judgement of others. We have included a clearer statement of the structure of this type of task, and more clearly cited the origin for its structure (Murphy & Ackerman, 2011):

'Our present work sought to achieve two primary goals. 1. Extend prior causal computational theories to formalise and test the interrelation between self-insertion and social contagion on learning and behaviour to better probe interpersonal generalisation in health, and 2., Test whether previous computational findings of social learning changes in BPD can be explained by infractions to self-other generalisation. We accomplish these goals by using a dynamic, sequential social value economic paradigm, the Intentions Game, building upon a Social Value Orientation Framework (Murphy & Ackerman, 2011) that assumes motivational variation in joint reward allocation.'

Given the introductions structure as it stands, we felt providing another paragraph on the normative assumptions of such a game was outside the scope of this article.

'The finding that individuals with BPD do not engage in self-other generalization on this task of social intentions is novel and potentially clinically relevant. The authors find that BPD participants' tendency to be prosocial when splitting points with a partner does not transfer into their expectations of how a partner will treat them in a task where they are the passive recipient of points chosen by the partner. In the discussion, the authors reasonably focus on model differences between groups (Bayesian model comparison), yet I thought this finding -- BPD participants not assuming prosocial tendencies in phase 2 while CON participant did -- merited greater attention. Although the BPD group was close to 0 on the \beta prior in Phase 2, their difference from CON is still in the direction

of being more mistrustful (or at least not assuming prosociality). This may line up with broader clinical literature on mistrustfulness and attributions of malevolence in the BPD literature (e.g., a 1992 paper by Nigg et al. in Journal of Abnormal Psychology). My broad point is to consider further the Phase 2 findings in terms of the clinical interpretation of the shift in β relative to controls.'

This is an important point, that we contextualize within the parameterisation of our utility model. While the shift toward 0 in the BPD participants is indeed more competitive, as the reviewer notes, it is surprisingly centred closely around 0, with only a slight bias to be prosocial (mean = -0.47; = -6.10, 95%HDI: -7.60, -4.60). Charitably we might argue that BPD participants are expecting more competitive preferences from their partner. However even so, given their variance around their priors in phase 2, they are uncertain or unconfident about this. We take a more conservative approach in the paper and say that given the tight proximity to 0 and the variance of their group priors, they are likely to be 'hedging their bets' on whether their partner is going to be prosocial or competitive. While the movement from phase 1 to 2 is indeed in the competitive direction it still lands in neutral territory. Model M4 does not preclude central tendencies at the start of Phase 2 being more in the competitive direction.

*'First, the authors note that they have "proposed a theory with testable predictions" (p. 4 but also elsewhere) but they do not state any clear predictions in the introduction, nor do they consider what sort of patterns will be observed in the BPD group in view of extant clinical and computational literature. Rather, the paper seems to be somewhat exploratory, largely looking at group differences (BPD vs. CON) on all of the shared computational parameters and additional indices such as belief updating and reaction times. Given this, I would suggest that the authors make stronger connections between extant research on intention representation in BPD and their framework (model and paradigm). In particular, the authors do not address related findings from Ereira (2020) and Story (2024) finding that in a false belief task that BPD participants *overgeneralize* from self to other. A critical comparison of this work to the present study, including an examination of the two tasks differ in the processes they measure, is important.'*

Thank you for this opportunity to include more of the important work that has preceded the present manuscript. Prior work has tended to focus on either descriptive explanations of self-other generalisation (e.g. through the use of RW type models) or has focused on observational learning instability in absence of a causal model from where initial self-other beliefs may arise. While the prior work cited by the reviewer [Ereira (2020; Nat. Comms.) and Story (2024; Trans. Psych.)] does examine the inter-trial updating between self-other, it does not integrate a self model into a self's belief about an other prior to observation. Rather, it focuses almost exclusively on prediction error 'leakage' generated during learning about individual reward (i.e. one sided reward). These findings are important, but lie in a slightly different domain. They also do not cut against ours, and in fact, we argue in the discussion that the sort of learning instability described above and splitting (as we cite from Story et al. 2024; Psych. Rev.) may result from a lack of self anchoring typical of CON participants. Nevertheless we agree these works provide an important premise to contrast and set the groundwork for our present analysis and have included them in the framing of our introduction, as well as contrasting them to our data in the discussion.

In the introduction:

'The BPD phenotype has been associated with a potential over-reliance on social versus internal cues (Henco et al., 2020), 'splitting' of social latent states that encode beliefs about others (Story et al., 2023), negative appraisal of interpersonal experiences with heightened self-blame (Mancinelli et al., 2024), inaccurate inferences about others' irritability (Hula et al., 2018), and reduced belief adaptation in social learning contexts (Siegel et al., 2020).

Associative models have also been adapted to characterize ‘leaky’ self-other reinforcement learning (Ereira et al., 2018), finding that those with BPD overgeneralize (leak updates) about themselves to others (Story et al., 2024). Altogether, there is currently a gap in the direct causal link between insertion, contagion, and learning (in)stability.’

In the discussion:

‘Disruptions in self-to-other generalization provide an explanation for previous computational findings related to task-based mentalizing in BPD. Studies tracking observational mentalizing reveal that individuals with BPD, compared to those without, place greater emphasis on social over internal reward cues when learning (Henco et al., 2020; Fineberg et al., 2018). Those with BPD have been shown to exhibit reduced belief adaptation (Siegel et al., 2020) along with ‘splitting’ of latent social representations (Story et al., 2024a). BPD is also shown to be associated with overgeneralisation in self-to-other belief updates about individual outcomes when using a one-sided reward structure (where participant responses had no bearing on outcomes for the partner; Story et al., 2024b). Our analyses show that those with BPD are equal to controls in their generalisation of absolute reward (outcomes that only affect one player) but disintegrate beliefs about relative reward (outcomes that affect both players) through adoption of a new, neutral belief. We interpret this together in two ways: 1. There is a strong concern about social relativity when those with BPD form beliefs about others, 2. The absence of constrained self-insertion about relative outcomes may predispose to brittle or ‘split’ beliefs. In other words, those with BPD assume ambiguity about the social relativity preferences of another (i.e. how prosocial or punitive) and are quicker to settle on an explanation to resolve this. Although self-insertion may be counter-intuitive to rational belief formation, it has important implications for sustaining adaptive, trusting social bonds via information moderation.’

In addition, perhaps it is fairer to note more explicitly the exploratory nature of this work. Although the analyses are thorough, many of them are not argued for a priori (e.g., rate of belief updating in Figure 2C) and the reader amasses many individual findings that need to be synthesized.’

We have now noted the primary goals of our work in the introduction, and have included caveats about the exploratory nature of our analyses. We would note that our model is in effect a causal combination of prior work cited within the introduction (Barnby et al., 2022; Moutoussis et al., 2016). This renders our computational models in effect a causal theory to test, although we agree that our dissection of the results are exploratory. We have more clearly signposted this:

‘Our present work sought to achieve two primary goals. 1. Extend prior causal computational theories to formalise and test the interrelation between self-insertion and social contagion on learning and behaviour to better probe interpersonal generalisation in health, and 2., Test whether previous computational findings of social learning changes in BPD can be explained by infractions to self-other generalisation. We accomplish these goals by using a dynamic, sequential economic paradigm, the Intentions Game, building upon a Social Value Orientation Framework (Murphy & Ackerman, 2011) that assumes innate motivational variation in joint reward allocation.’

‘Second, in the discussion, the authors are too quick to generalize to broad clinical phenomena in BPD that are not directly connected to the task at hand. For example, on p. 22: “Those with a diagnosis of BPD also show reduced permeability in generalising from other to self. While prior research has predominantly focused on how those with BPD use information to form impressions, it has not typically examined whether these impressions affect the self.” Here, it’s not self-representation per se (typically, identity or one’s view of oneself), but instead cooperation and prosocial tendencies in an economic

context. It is important to clarify what clinical phenomena may be closely related to the task and which are more distal and perhaps should not be approached here.'

Thank you for this important point. We agree that social value orientation, and particularly in this economically-assessed form, is but one aspect of the self, and we did not test any others. A version of the social contagion phenomena is also present in other aspects of the self in intertemporal (Moutoussis et al., 2016), economic (Suzuki et al., 2016) and moral preferences (Yu et al., 2021). It would be most interesting to attempt to correlate the degrees of insertion and contagion across the different tasks.

We take seriously the wider concern that behaviour in our tasks based on economic preferences may not have clinical validity. This issue is central in the whole field of computational psychiatry, much of which is based on generalizing from tasks like ours, and discussing correlations with psychometric measures. We hope that it is acceptable to leave such discussions to the many reviews on computational psychiatry (Montague et al., 2012; Hitchcock et al., 2022; Huys et al., 2016). Here, we have just put a caveat in the discussion:

'Finally, a limitation may be that behaviour in tasks based on economic preferences may not have clinical validity. This issue is central to the field of computational psychiatry, much of which is based on generalising from tasks like that within this paper and discussing correlations with psychometric measures. Extrapolating economic tasks into the real world has been the topic of discussion for the many reviews on computational psychiatry (e.g. Montague et al., 2012; Hitchcock et al., 2022; Huys et al., 2016). We note a strength of this work is the use of model comparison to understand causal algorithmic differences between those with BPD and matched healthy controls. Nevertheless, we wish to further pursue how latent characteristics captured in our models may directly relate to real-world affective change.'

'On a more technical level, I had two primary concerns. First, although the authors consider alternative models within a hierarchical Bayesian framework, some challenges arise when one analyzes parameter estimates fit separately to two groups, particularly when the best-fitting model is not shared. In particular, although the authors conduct a model confusion analysis, they do not as far I could tell (and apologies if I missed it) demonstrate that the dynamics of one model are nested within the other. Given that M4 has free parameters governing the expectations on the absolute and relative reward preferences in Phase 2, is it necessarily the case that the shared parameters between M1 and M4 can be interpreted on the same scale? Relatedly, group-specific model fitting has virtues when believes there to be two distinct populations, but there is also a risk of overfitting potentially irrelevant sample characteristics when parameters are fit group by group.

*To resolve these issues, I saw one straightforward solution (though in modeling, my experience is that what seems straightforward on first glance may not be so upon further investigation). M1 assumes that participants' own preferences (posterior central tendency) in Phase 1 directly transfer to priors in Phase 2, but presumably the degree of transfer could vary somewhat without meriting an entirely new model (i.e., the authors currently place this question in terms of model selection, not within-model parameter variation). I would suggest that the authors consider a model parameterization fit to the full dataset (both groups) that contains free parameters capturing the *deviations* in the priors relative to the preceding phase's posterior. That is, the free parameters $\bar{\alpha}_{par}^m$ and $\bar{\beta}_{par}^m$ govern the central tendency of the Phase 2 prior parameter distributions directly, but could be reparametrized as deviations from Phase 1 θ^m_{ppt} parameters in an additive form. This allows for a single model to be fit all participants that encompasses the dynamics of interest such that between-group parameter comparisons are not biased by the strong assumptions*

imposed by M1 (that phase 1 preferences and phase 2 observations directly transfer to priors). In the case of controls, we would expect these deviation parameters to be centred on 0 insofar as the current M1 fit them best, whereas for BPD participants should have significant deviations from earlier-phase posteriors (e.g., the shift in β toward prior neutrality in phase 2 compared to one's own prosociality in phase 1). I think it's still valid for the authors to argue for stronger model constraints for Bayesian model comparison, as they do now, but inferences regarding parameter estimates should ideally be based on a model that can encompass the full dynamics of the entire sample, with simpler dynamics (like posterior $\rightarrow$ prior transfer) being captured by near-zero parameter estimates.'

Thank you for the chance to be clearer in our modelling. In particular, the suggestion to include a model that can be fit to all participants with the equivalent of the likes of partial social insertion, to check if the results stand, can actually be accomplished through our existing models. That is, the θ_{nar}^{ref} parameter that governs the flexibility over beliefs in phase 2 under models M1 (dominant for CON participant) and M2 parameterises the degree to which participants think their partner may be different from themselves. Thus, forcibly fitting M1 and M2 hierarchically to all participants, and then separately to BPD and CON participants, can quantify the issue raised: if BPD participants indeed distinguish partners as vastly different from themselves enough to warrant a new central tendency, θ_{nar}^{ref} should be quantitatively higher in BPD vs CON participants under M1 and M2.

We therefore tested this, reporting the distributional differences between θ_{nar}^{ref} for BPD and CON participants under M1, both when fitted together as a population and as separate groups. As θ_{nar}^{ref} is higher for BPD participants under both conditions for M1 and M2 it supports our claim and will add more context for the comparison - β_{par}^{ref} may be large enough in BPD that a new central tendency to anchor beliefs is a more parsimonious explanation.

We cross checked this result by assessing the discrepancy between the participant's and assumed partner's central tendencies for both prosocial and individualistic preferences via best-fitting model M4 for the BPD group. We thereby examined whether belief disintegration is uniform across preferences (relative vs absolute reward) or whether one tendency was shifted dramatically more than another. We found that beliefs over prosocial-competitive preferences were dramatically shifted, whereas those over individualistic preferences were not.

We have added the following to the main text results to explain this:

Model Comparison:

'We found that CON participants were best fit at the group level by M1 (Frequency = 0.59, Protected Exceedance Probability = 0.98), whereas BPD participants were best fit by M4 (Frequency = 0.54, Protected Exceedance Probability = 0.86; Figure 2A). We first analyse the results of these separate fits. Later, in order to assuage concerns about drawing inferences from different models, we examined the relationships between the relevant parameters when we forced all participants to be fit to each of the models (in a hierarchical manner, separated by group). In sum, our model comparison is supported by convergence in

parameter values when comparisons are meaningful. We refer to both types of analysis below.’

Phase 1:

‘These differences were replicated when considering parameters between groups when we fit all participants to the same models (M1-M4; see Table S2).’

Phase 2:

‘To check that these conclusions about self-insertion did not depend on the different models, we found that only β_{par}^{ref} under M1 and M2 were consistently larger in BPD versus CON. This

supports the notion that new central tendencies for BPD participants in phase 2 were required, driven by expectations about a partner’s relative reward. (see Fig S10 & Table S2).

α_{par}^{ref} and β_{par}^{ref} parameters under assumptions of M1 and M2 were strongly correlated with

median change in belief between phase 1 and 2 under M3 and M4, suggesting convergence in outcome (Fig S11).’

‘Furthermore, even under assumptions of M1-M4 for both groups, BPD showed smaller posterior median changes versus CON in phase 2 (see Table T2). These results converge to suggest those with BPD form rigid posterior beliefs.’

‘Assessing this same relationship under M1- and M2-only assumptions reveals a replication of this group effect for absolute reward, but the effect is reversed for relative reward (see Table S3). This accords with the context of each model, where under M1 and M2, BPD participants had larger phase 2 prior flexibility over relative reward (leading to larger initial surprise), which was better accounted for by a new central tendency under M4 during model comparison. When comparing both groups under M1-M4 informational surprise over absolute reward was consistently restricted in BPD (Table S3), suggesting a diminished weight of this preference when forming beliefs about an other.’

Phase 3

‘In the dominant model for the BPD group—M4—participants are not influenced in their phase 3 choices following exposure to their partner in phase 2. To further confirm this we also analysed absolute change in median participant beliefs between phase 1 and 3 under the assumption that M1 and M3 was the dominant model for both groups (that allow for contagion to occur). This analysis aligns with our primary model comparison using M1 for CON and M4 for BPD (Figure 2C). CON participants altered their median beliefs between phase 1 and 3 more than BPD participants (M1: linear estimate = 0.67, 95%CI: 0.16, 1.19; $t = 2.57$, $p = 0.011$; M3: linear estimate = 1.75, 95%CI: 0.73, 2.79; $t = 3.36$, $p < 0.001$). Relative reward was overall more susceptible to contagion versus absolute reward (M1: linear estimate = 1.40, 95%CI: 0.88, 1.92; $t = 5.34$, $p < 0.001$; M3: linear estimate = 2.60, 95%CI: 1.57, 3.63; $t = 4.98$, $p < 0.001$). There was an interaction between group and belief type under M3 but not M1 (M3: linear estimate = 2.13, 95%CI: 0.09, 4.18, $t = 2.06$, $p = 0.041$). There was only a main effect of belief type on precision under M3 (linear estimate = 0.47, 95%CI: 0.07, 0.87, $t = 2.34$, $p = 0.02$); relative reward preferences became more precise across the board. Derived model estimates of preference change between phase 1 and 3 strongly correlated between M1 and M3 along both belief types (see Table S2 and Fig S11).’

‘My second concern pertains to the psychometric individual difference analyses. These were not clearly justified in the introduction, though I agree that they could offer potentially meaningful insight into which scales may be most related to model parameters of interest. So, perhaps these should be earmarked as exploratory and/or

more clearly argued for. Crucially, however, these analyses appear to have been conducted on the full sample without considering the group structure. Indeed, many of the scales on which there are sizable group differences are also those that show correlations with psychometric scales. So, in essence, it is unclear whether most of these analyses are simply recapitulating the between-group tests reported earlier in the paper or offer additional insights. I think it's hard to have one's cake and eat it, too, in this regard and would suggest the authors review Preacher et al. 2005, Psychological Methods for additional detail. One solution might be to always include group as a binary covariate in the symptom dimension-parameter analyses, essentially partialing the correlations for group status. I remain skeptical regarding whether there is additional signal in these analyses, but such controls could convince the reader. Nevertheless, without such adjustments, I would caution against any transdiagnostic interpretations such as this one in the Highlights: "Higher reported childhood trauma, paranoia, and poorer trait mentalizing all diminish other-to-self information transfer irrespective of diagnosis." Since many of these analyses relate to scales on which the groups differ, the transdiagnostic relevance remains to be demonstrated.'

We have restructured the psychometric section to ensure transparency and clarity in our analysis. Namely, in response to these comments and those of the other reviewers, we have opted to remove the parameter analyses that aimed to cross-correlate psychometric scores with latent parameters from different models: as the reviewer points out, we do not have parity between dominant models for each group to warrant this, and fitting the same model to both groups artificially makes the parameters qualitatively different. Instead we have

opted to focus on social contagion, or rather restrictions on $\Delta\beta_{ppt}^m$, $\Delta\alpha_{ppt}^m$ between phases 1 and

3 explained by M3. This provides us with an opportunity to examine social contagion on the whole population level isolated from self-insertion biases. We performed bootstrapping (1000 reps) and permutation testing (1000 reps) to assess the stability and significance of each edge in the partial correlation network, and then applied FDR correction (p[fdr]), thus controlling for multiple comparisons. We note that while we focused on M3 to isolate the effect across the population, social contagion across both relative and absolute reward under M3 strongly correlated with social contagion under M1 (see Fig S11).

'We explored whether social contagion may be restricted as a result of trauma, paranoia, and less effective trait mentalizing under the assumption of M3 for all participants (where everyone is able to be influenced by their partner). To note, social contagion under M3 was highly correlated with contagion under M1 (see Fig S11). We conducted partial correlation analysis to estimate relationships conditional on all other associations and retained all that survived bootstrapping (1000 reps), permutation testing (1000 reps), and subsequent FDR correction. Persecution and CTQ scores were both moderately associated with MZQ scores (RGPTSB $r = 0.41$, 95%CI: 0.23, 0.60, $p = 0.004$, p[fdr]=0.043; CTQ $r = 0.354$ 95%CI: 0.13, 0.56, $p=0.019$, p[fdr]=0.02). MZQ scores were in turn moderately and negatively associated with

shifts in prosocial-competitive preferences ($\Delta\beta_{ppt}^m$) between phase 1 and 3 ($r = -0.26$, 95%CI:

-0.46, -0.06, $p=0.026$, p[fdr]=0.043). CTQ scores were also directly and negatively associated with shifts in individualistic preferences (; $r = -0.24$, 95%CI: -0.44, -0.13, $p=0.052$, p[fdr]=0.065). This provides some preliminary evidence that trauma impacts beliefs about individualism directly, whereas trauma and persecutory beliefs impact beliefs about prosociality through impaired mentalising (Figure 4A).'

(1) As far as I could tell, the authors didn't provide an explanation of this finding on page 5: "However, CON participants made significantly fewer prosocial choices when individualistic choices were available" While one shouldn't be forced to interpret every

finding, the paper is already in that direction and I found this finding to be potentially relevant to the BPD-control comparison.

Thank you for this observation. This sentence reports the fact that CON participants were effectively more selfish than BPD participants. This is captured by the lower value of α_{ppt}^m .

reported in Figure 2, and suggests that CON participants were more focused on absolute value – acting in a more ‘economically rational’ manner – versus BPD participants. This fits in with our fourth paragraph of the discussion where we discuss prior work that demonstrates a heightened social focus in those with BPD. Indeed, the finding the reviewer highlights further emphasises the point that those with BPD are much more sensitive, and motivated to choose, options concerning relative reward than are CON participants. The text in the discussion reads:

‘We also observe this in self-generated participant choice behaviour, where CON participants were more concerned over absolute reward versus their BPD counterparts, suggesting a heightened focus on relative vs. absolute reward in those with BPD.’

(2) The adaptive algorithm for adjusting partner behavior in Phase 2 was clever and effective. Did the authors conduct a manipulation check to demonstrate that the matching resulted in approximately 50% difference between one's behavior in Phase 1 and the partner in Phase 2? Perhaps Supplementary Figure suffices, but I wondered about a simpler metric.

Thanks for this point. We highlight this in Figure 3B and within the same figure legend although appreciate the panel is quite small and may be missed. We have now highlighted this manipulation check more clearly in behavioural analysis section of the main text:

‘Server matching between participant and partner in phase 2 was successful, with participants being approximately 50% different to their partners with respect to the choices each would have made on each trial in phase 2 (mean similarity=0.49, SD=0.12).’

(3) The resolution of point-range plots in Figure 4 was grainy. Perhaps it's not so in the separate figure file, but I'd suggest checking.

Apologies. We have now updated and reorganised the figure to improve clarity.

(4) p. 21: Suggest changing to "different" as opposed to "opposite" since the strategies are not truly opposing: "but employed opposite strategies."

We have amended this.

(5) p. 21: I found this sentence unclear, particularly the idea of "similar updating regime." I'd suggest clarifying: "In phase 2, CON participants exhibited greater belief sensitivity to new information during observational learning, eventually adopting a similar updating regime to those with BPD."

We have clarified this statement:

‘In observational learning in phase 2, CON participants initially updated their beliefs in response to new information more quickly than those with BPD, but eventually converged to a similar rate of updating.’

(6) p. 23: The content regarding psychosis seemed out of place, particularly as the concluding remark. I'd suggest keeping the focus on the clinical population under

investigation. If you'd like to mention the paradigm's relevance to psychosis (which I think could be omitted), perhaps include this as a future direction when describing the paradigm's strengths above.

We agree the paragraph is somewhat speculative. We have omitted it in aid of keeping the messaging succinct and to the point.

(7) p. 24: Was BPD diagnosis assess using unstructured clinical interview? Although psychosis was exclusionary, what about recent manic or hypomanic episodes or Bipolar diagnosis? A bit more detail about BPD sample ascertainment would be useful, including any instruments used to make a diagnosis and information about whether you measured inter-rater agreement.

Participants diagnosed with BPD were recruited from specialist personality disorder services across various London NHS mental health trusts. The diagnosis of BPD was established by trained assessors at the clinical services and confirmed using the Structured Clinical Interview for DSM-IV (SCID-II) (First et al., 1997). Individuals with a history of psychotic episodes, severe learning disability or neurological illness/trauma were excluded. We have now included this extra detail within our methods in the paper:

‘The majority of BPD participants were recruited through referrals by psychiatrists, psychotherapists, and trainee clinical psychologists within personality disorder services across 9 NHS Foundation Trusts in the London, and 3 NHS Foundation Trusts across England (Devon, Merseyside, Cambridgeshire). Four BPD participants were also recruited by self-referral through the UCLH website, where the study was advertised. To be included in the study, all participants needed to have, or meet criteria for, a primary diagnosis of BPD (or emotionally-unstable personality disorder or complex emotional needs) based on a professional clinical assessment conducted by the referring NHS trust (for self-referrals, the presence of a recent diagnosis was ascertained through thorough discussion with the participant, whereby two of the four also provided clinical notes). The patient participants also had to be under the care of the referring trust or have a general practitioner whose details they were willing to provide. Individuals with psychotic or mood disorders, recent acute psychotic episodes, severe learning disability, or current or past neurological disorders were not eligible for participation and were therefore not referred by the clinical trusts.’

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